Grouped recursion and colour registers: an improved exponent for the exact power-saving DFT

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October 8, 2026

Abstract

OpenAI recently gave a deterministic algorithm that computes the discrete Fourier transform of every length n exactly in $O\big(n(\log n)^{\theta}(\log\log n)^{4-\theta}\big)$ operations, with $1-\theta \approx 2.1\cdot 10^{-13}$, and a Lean formalization of a variant with $1-\theta = 2\cdot 10^{-11}$. Their method converts a saving for one finite binary "network" into a fast algorithm for the tensor powers $C^{\otimes k}$ of a 2×2 matrix, by a recursion that treats every directional factor of the network as a separate recursive call. We make two improvements. First, the *grouped recursion*: all directional factors of one edge of the network form a single tensor power and can be run as one recursive call on a larger fibre. The exponent becomes the root of $\sum_e (\rho_e/m)^p = W^*$ instead of $\log_m(\sum_e \rho_e/W^*)$. Second, *colour registers*: a new network in which most of the paper's side wires are replaced by a small number of shared registers. Together they give the exact DFT in

$$O\Big(n\,(\log n)^{\,1-3.2\cdot 10^{-6}}\Big)$$

operations (Theorem 1.1). The grouped recursion alone, applied to the network that OpenAI formalized, gives $O\big(n(\log n)^{1-10^{-8}}\big)$ (Theorem 1.2). Both results have been formally verified in the proof assistant Lean 4, as an extension of OpenAI's formalization.

Contents

1 Introduction **2**
- 1.1 Background . . . 2
- 1.2 Results . . . 2
- 1.3 Ideas . . . 3
- 1.4 Organization . . . 3

2 Preliminaries **4**
- 2.1 Model and reduction . . . 4
- 2.2 Binary labels and frames . . . 4
- 2.3 Network certificates . . . 4

3 The grouped recursion **5**
- 3.1 One edge is one tensor power . . . 5
- 3.2 The grouped work function . . . 6
- 3.3 The grouped tensor-power theorem . . . 6
- 3.4 Certifying the rate inequality . . . 8

4 OpenAI's formalized network **8**

5 Colour registers **9**
5.1 Factor data and centres . . . 10
5.2 Meeting labels . . . 11
5.3 Register designs . . . 12
5.4 The invocations . . . 13
5.5 Labels and edges . . . 15
5.6 The network . . . 17
5.7 Designs . . . 19
5.8 The exponent . . . 22
5.9 The colour-register theorem . . . 24

6 Proof of the Main Theorem **25**

7 Formal verification **25**

8 Summary and outlook **26**

1 Introduction

1.1 Background

For $n \geq 1$ let $F_n = (\zeta_n^{jk})_{0 \leq j,k < n}$ with $\zeta_n = e^{2\pi i/n}$. The fast Fourier transform computes $F_n x$ in $O(n \log n)$ arithmetic operations at every length. OpenAI [1] recently gave the first power saving in the logarithm, in a model of exact complex arithmetic with specified roots, unrestricted coefficients and unit-cost logarithmic-size indexing, in which all scalar preparation and array organization are charged:

Theorem (OpenAI [1, Theorem 1.1])**.** *There is a single deterministic algorithm which, for every $n \geq 1$ and $x \in \mathbb{C}^n$, computes $F_n x$ exactly in $O\big(n(\log n)^{\theta}(\log\log n)^{4-\theta}\big)$ operations, where $\theta = \log(10^6 - \Delta/2^{71})/\log 10^6$ with $\Delta = 6\,871\,402\,692\,000\,000$; thus $1 - \theta \approx 2.1 \cdot 10^{-13}$.*

The proof has three layers. A finite combinatorial object, a *network* on m binary coordinates with W^* roles, yields an algorithm for the tensor powers $C^{\otimes k}$ of $C = \frac{1}{2}\left(\begin{smallmatrix} 1+i & 1-i \\ 1-i & 1+i \end{smallmatrix}\right)$ [1, Theorem 2.6]. The exponent of that algorithm then propagates through a reduction of F_n to such tensor powers [1, Sections 3–5]. OpenAI also formalized the whole argument in Lean 4 [2], for a smaller explicit network, proving the internal exponent $1 - 2 \cdot 10^{-11}$.

1.2 Results

Our main result is the following. It has been formally verified (Section 7).

Theorem 1.1 (Main Theorem)**.** *There is a single deterministic algorithm which, for every $n \geq 1$ and $x \in \mathbb{C}^n$, computes $F_n x$ exactly in the model of [1] in*

$$O(n\,(\log n)^{q})\,, \qquad q = 1 - 3.2 \cdot 10^{-6},$$

operations. More precisely, the bound $O\big(n(\log n)^{z}(\log\log n)^{4-z}\big)$ holds with $z = 1 - 3.21 \cdot 10^{-6}$; the exact root of the rate equation of the network is $1 - p = 3.2791 \cdot 10^{-6}$. The same holds for exact convolution of two length-n sequences.

The grouped recursion alone, on the network formalized by OpenAI, already gives a substantial improvement over the exponent of [2]. This secondary result has also been formally verified.

Theorem 1.2 (grouped recursion on OpenAI's formalized network)**.** *In the RAM model and with the correctness specification of OpenAI's formalization [2], there is a fixed pair of programs computing $F_n x$ exactly for every $n \geq 1$ and $x \in \mathbb{C}^n$, with work*

$$W(n) = O\Big(n\,(\log n)^{1-10^{-8}}\Big)\,.$$

The same holds for exact convolution.

Remark 1.3 (formal verification). Theorems 1.1 and 1.2 have been formally verified in Lean 4 with the Mathlib library, as an extension of OpenAI's formalization [2], using only the standard axioms and no unverified computation. The formal proof of Theorem 1.1 certifies the rate inequality at $z = 1 - 3.21 \cdot 10^{-6}$ and obtains the slightly stronger bound $O\big(n(\log n)^z(\log\log n)^2\big)$. See Section 7.

source	network	recursion	exponent gap $1-q$
[1, Thm. 1.1]	$h = 100$, $m = 10^6$, $W^* = 2^{71}$	per direction	$2.1 \cdot 10^{-13}$
OpenAI's formal proof [2]	32 coordinates per factor, $W^* = 2^{50}$	per direction	$2 \cdot 10^{-11}$
Theorem 1.2 (formally verified)	the same	grouped	10^{-8} (internal $1.4657 \cdot 10^{-8}$)
Theorem 1.1 (formally verified)	colour registers, $(30, 20, 41)$, $W^* = 2^{43}$	grouped	$3.2 \cdot 10^{-6}$ (internal $3.279 \cdot 10^{-6}$)

1.3 Ideas

Grouped recursion. In [1], an edge of the network along which a role's label grows by ρ binary dimensions costs ρ *directional kernels* C_z. Each of these is run as separate recursive calls on fibres of size $\lfloor k/m \rfloor$. The ρ kernels of one edge act along an orthonormal, hence linearly independent, family of directions. After one change of coordinates their product is $C^{\otimes\rho}$ on ρ coordinates (Lemma 3.1). Hence the whole edge is one batch of recursive calls on fibres of size $\rho\lfloor k/m \rfloor$. The cost recurrence becomes $t(k) \leq A + (W^*)^{-1} \sum_e t(\rho_e \lfloor k/m \rfloor)$, with exponent the root p of $\sum_e (\rho_e/m)^p = W^*$ (Theorem 3.7). This p is never larger than the paper's exponent (Lemma 3.9). It is much smaller when most of the total dimension sits in edges of size close to m, which is the case for these networks: on OpenAI's formalized network, grouping improves $1 - \theta$ from $7.7 \cdot 10^{-11}$ to $1.47 \cdot 10^{-8}$.

Colour registers. The paper's network spends one auxiliary wire, a *side wire*, for every ordered pair of orthogonal triples, i.e. $\binom{h-3}{3} + 3(h-3)$ wires per element (8550 at $h = 41$). In our network each element owns a few *registers*, which meet the registers of other elements at shared *meeting labels*. A catalytic 12-step schedule realizes the required transfer matrix for arbitrary auxiliary contents. With searched register designs this costs about 73 auxiliary wires per element at $h = 41$ (Section 5).

1.4 Organization

Section 2 recalls the model, labels, frames, networks and the reduction to tensor powers. Section 3 proves the grouped tensor-power theorem. Section 4 applies it to OpenAI's formalized network and proves Theorem 1.2. Section 5 constructs the colour-register network, and Section 6 completes the proof of Theorem 1.1. Section 7 describes the formal verification, and Section 8 summarizes and lists open problems.

2 Preliminaries

We use the notation of [1] throughout.

2.1 Model and reduction

The model of computation is that of [1, Section 1]: exact complex registers with unit-cost field operations, rational constants and i, specified roots of unity as inputs, and unit-cost integer and address operations on words of $O(\log n)$ bits. All scalar preparation, table construction and index work are charged. In OpenAI's formalization [2] this model is a first-order RAM language with an operation-count semantics. The reduction of the DFT to tensor powers is the following.

Proposition 2.1 ([1, Sections 3–5])**.** *Let $0 < q \le 1$. Suppose there is a deterministic exact algorithm applying $C^{\otimes k}$ to an arbitrary complex array of length 2^k in $O(2^k(k+1)^q)$ operations for every $k \ge 0$, including fixed-table construction, scalar arithmetic, index preparation and data movement. Then $F_n x$ can be computed for every $n \ge 1$ in $O\big(n(\log n)^q(\log\log n)^{4-q}\big)$ operations. The same holds for exact convolution.*

Indeed, the proof of [1, Theorem 1.1] uses [1, Theorem 2.6] only through the bound $O(2^k(k+1)^\theta)$ with $0 < \theta \le 1$: in the sector estimate of [1, Proposition 4.2] and in [1, Section 5]. In the formalization [2] the argument downstream of the tensor-power bound uses the exponent only through $0 < q < 1$, and proves the slightly stronger bound $O\big(n(\log n)^q(\log\log n)^2\big)$.

Corollary 2.2. *Under the hypothesis of Proposition 2.1, F_n costs $O(n(\log n)^{q'})$ for every $q' > q$.*

Proof. $(\log n)^q(\log\log n)^{4-q} = o\big((\log n)^{q'}\big)$. □

2.2 Binary labels and frames

Let $\mathbb{D} = \mathbb{F}_2^m$ with the dot product $x \cdot y = \sum_i x_i y_i$. A subspace $U \le \mathbb{D}$ is *nondegenerate* if the restricted form has zero radical. For nested nondegenerate $U \subseteq V$ the *residual* is $V \cap U^\perp$. A family $z_1, \dots, z_\rho$ is *orthonormal* if $z_j \cdot z_k = \delta_{jk}$; an orthonormal family is linearly independent, since its Gram matrix is the identity. For $z \in \mathbb{D}$ let R_z be the translation $(R_z g)(x) = g(x+z)$ of functions $g : \mathbb{D} \to \mathbb{C}$. Let $a = \frac{1+i}{2}$, $b = \frac{1-i}{2}$, and let

$$C_z = aI + bR_z$$

be the *directional kernel* along z. For a coordinate vector e_j, C_{e_j} is C acting on coordinate j. For a nondegenerate U the frame Φ_U is defined in [1, (2.5)]; we use only the following property.

Lemma 2.3 ([1, Lemma 2.2])**.** *Let $U \subseteq V$ be nested nondegenerate subspaces whose residual has an orthonormal basis $z_1, \dots, z_\rho$. Then $\Phi_V \Phi_U^{-1} = R_\tau \prod_{j=1}^{\rho} C_{z_j}$ for some $\tau \in \mathbb{D}$, and $\Phi_U \Phi_V^{-1} = R_{\tau'} \prod_j C_{z_j}$ for some τ'.*

(The statement in [1] speaks of ρ factors $C_z^{\pm 1}$; since $C_z^{-1} = R_z C_z$ and all R_z, C_w commute, the translations collect into one R_τ.)

2.3 Network certificates

We isolate the interface between a network and the recursion. Let $W^* \ge 1$ and $f \ge 1$, and consider W^* complex arrays (*roles*), each indexed by $(\mathbb{F}_2^m)^f$, i.e. by f *columns* of m bits. A matrix acting on functions $\mathbb{F}_2^m \to \mathbb{C}$ acts on such an array by its f-fold tensor power.

Definition 2.4 (network certificate)**.** An (m, W^*)-*certificate* is a finite word of moves of the following three kinds.

1. A *gate* $g \in \mathbb{Q}^{W^* \times W^*}$: it replaces the roles $(x_r)_r$ by $(\sum_s g_{rs} x_s)_r$, pointwise in the address.
2. A *translation* (r, τ): role r is multiplied by R_τ.
3. An *edge move* $(r, \tau, z_1, \dots, z_\rho)$ with $z_1, \dots, z_\rho$ orthonormal: role r is multiplied by $R_\tau \prod_{j=1}^{\rho} C_{z_j}$. Its *size* is ρ.

The word must act as $C^{\otimes mf}$ on every role, for every $f \geq 1$ and arbitrary contents. Its *edge multiset* $\mathcal{E}$ is the multiset of sizes of its edge moves.

Remark 2.5. Every network of [1, Section 2] gives a certificate. Its gates are the scalar gates. Its edge moves are the frame changes $\Phi_V \Phi_U^{-1}$ between consecutive labels of a role, by Lemma 2.3, with the orthonormal basis of the residual. Its terminal corrections are translations and a signed permutation gate. This is [1, Proposition 2.4] together with [1, (2.6)]. A role that the network does not use (a *padding role*) receives the path $0 \to \langle e_1 \rangle \to \mathbb{D}$, i.e. two edges of sizes 1 and $m-1$. Its frame ratio is $\Phi_{\mathbb{D}} = C^{\otimes m}$.

Theorem 2.6 ([1, Theorem 2.6], per-direction recursion)**.** *Given an* (m, W^*)*-certificate with* $s = \sum_{e \in \mathcal{E}} \rho_e < W^* m$*, the tensor power* $C^{\otimes k}$ *can be applied in* $O(2^k (k+1)^\theta)$ *operations,* $\theta = \log_m(s/W^*)$.

The proof of [1] runs every factor C_{z_j} of an edge move as 2^{k-f} copies of $C^{\otimes f}$ on fibres, $f = \lfloor k/m \rfloor$, grouped into $2^{k-f}/W^*$ recursive calls of the W^*-role algorithm. It thus obeys $t(k) \leq A + \frac{s}{W^*} t(\lfloor k/m \rfloor)$ for $t(k) = T(k)/(W^* 2^k)$.

3 The grouped recursion

3.1 One edge is one tensor power

For $M \in GL_m(\mathbb{F}_2)$ let $(P_M g)(x) = g(M^{-1}x)$.

Lemma 3.1 (batch identity)**.** *Let* $z_1, \dots, z_\rho \in \mathbb{F}_2^m$ *be linearly independent and let* $M \in GL_m(\mathbb{F}_2)$ *satisfy* $M z_j = e_j$ $(1 \leq j \leq \rho)$*. Then*

$$\prod_{j=1}^{\rho} C_{z_j} = P_M^{-1} \big(C^{\otimes \rho} \otimes I_{2^{m-\rho}} \big) P_M .$$

On f *columns, with* M *applied in every column,* $\big(\prod_j C_{z_j}\big)^{\otimes f}$ *is a fixed address permutation, followed by* $2^{(m-\rho)f}$ *independent copies of* $C^{\otimes \rho f}$ *on contiguous fibres, followed by the inverse permutation.*

Proof. $(P_M R_z P_M^{-1} g)(x) = (P_M^{-1} g)(M^{-1}x + z) = g(x + Mz)$, so $P_M R_z P_M^{-1} = R_{Mz}$ and $P_M C_z P_M^{-1} = C_{Mz}$. Hence $\prod_j C_{z_j} = P_M^{-1} \big(\prod_j C_{e_j} \big) P_M$, and $C_{e_j} = aI + bR_{e_j}$ is C on coordinate j tensored with the identity. For the second statement tensor over the f columns, and order the addresses so that the ρf selected coordinates are the lowest ones. □

Remark 3.2. If the z_j are members v_i $(i \in R)$ of one orthonormal basis $v_1, \dots, v_m$ of $\mathbb{F}_2^m$, one may take $Mx = (v_i \cdot x)_i$, reordered so that the coordinates in R come last. Then $M^{-1} y = \sum_i y_i v_i$ by orthonormality.

Lemma 3.3 (batch addressing)**.** *Fix m and $M \in GL_m(\mathbb{F}_2)$. On an array of length 2^{mf+r} indexed by f columns and r spectator bits, the permutation of Lemma 3.1 and its inverse can be applied in $O(2^{mf+r})$ operations, with constants depending only on m.*

Proof. This is the proof of [1, Lemma 2.5], with "ρ selected output bits per column" in place of one. Tabulate once, for every m-bit column digit u, the value Mu, split into its ρ selected bits and its $m-\rho$ remaining bits; the table has $2^m = O(1)$ entries. Traverse the digit prefix tree of the address and add the two tabulated contributions per column, shifted to their output positions: $O(1)$ work per node and fewer than 2^{mf+r+1} nodes. For the inverse, exchange source and destination. □

Remark 3.4. The RAM model of [2] stores the f bits of each coordinate as one word (transposed digits). There the address map is computed coordinatewise as XOR-folds of words, in constant work per address.

3.2 The grouped work function

Definition 3.5 (grouped work function)**.** Let $m \geq 3$ and $a \geq 0$ be integers and $N : \{0, \dots, m-1\} \to \mathbb{N}$. Define $G : \mathbb{N} \to \mathbb{N}$ by

$$G(k) = \begin{cases} 2^k + \displaystyle\sum_{c=0}^{m-1} N(c)\, 2^{\,k-c\lfloor k/m\rfloor - a}\, G\big(c\lfloor k/m\rfloor\big) & \text{if } k \geq m(a+1), \\ 2^k & \text{otherwise.} \end{cases}$$

This is well defined: for $k \geq m(a+1)$ and $c < m$, $c\lfloor k/m\rfloor \leq (m-1)\lfloor k/m\rfloor < k$. Moreover $k - c\lfloor k/m\rfloor \geq \lfloor k/m\rfloor \geq a+1$, so the exponent $k - c\lfloor k/m\rfloor - a$ is a positive integer and

$$2^{\,k-c\lfloor k/m\rfloor - a} \cdot 2^a \cdot 2^{\,c\lfloor k/m\rfloor} = 2^k. \tag{1}$$

Lemma 3.6 (supersolution)**.** *In Definition 3.5 suppose $N(0) = 0$, and let $z \geq 0$ satisfy*

$$\Sigma := \sum_{c=1}^{m-1} N(c) \Big(\frac{c}{m}\Big)^z < 2^a.$$

Then $G(k) \leq K\, 2^k (k+1)^z$ for all $k \geq 0$, where $K = 1 + 2^a/(2^a - \Sigma)$.

Proof. We show $G(k) \leq K 2^k k^z$ for $k \geq 1$ by strong induction. If $k < m(a+1)$ then $G(k) = 2^k \leq K 2^k k^z$, since $K \geq 1$ and $k^z \geq 1$. Otherwise let $f = \lfloor k/m\rfloor \geq 1$. For $1 \leq c < m$ the child cf satisfies $1 \leq cf < k$ and $(cf)^z \leq (ck/m)^z = (c/m)^z k^z$. By induction and (1),

$$G(k) \leq 2^k + \sum_{c \geq 1} N(c)\, 2^{k-a}\, K \Big(\frac{c}{m}\Big)^z k^z = 2^k \Big(1 + K \frac{\Sigma}{2^a} k^z\Big) \leq K 2^k k^z,$$

where the last step uses $K(1 - \Sigma/2^a) = (1 - \Sigma/2^a) + 1 \geq 1$ and $k^z \geq 1$. Finally $G(0) = 1 \leq K$, and $k^z \leq (k+1)^z$. □

3.3 The grouped tensor-power theorem

Theorem 3.7 (grouped tensor-power theorem)**.** *Let an (m, W^*)-certificate be given with $m \geq 3$, $W^* = 2^a$, and edge multiset $\mathcal{E}$ with $1 \leq \rho \leq m-1$ for all $\rho \in \mathcal{E}$. Let $z \in [0, 1]$ satisfy*

$$\sum_{\rho \in \mathcal{E}} \Big(\frac{\rho}{m}\Big)^z < W^*. \tag{2}$$

Then there is a deterministic exact algorithm applying $C^{\otimes k}$ to an arbitrary complex array of length 2^k in $O\big(2^k(k+1)^z\big)$ operations, for every $k \geq 0$, in the model of Proposition 2.1.

Proof. We describe an algorithm $\mathcal{A}$ that transforms W^* arrays of length 2^k simultaneously. One array is handled by zero-filling the other $W^* - 1$.

Base range. If $k < K_0 := m(a+1)$, apply the k factors of C one at a time along their fibres, in $O(W^* k 2^k) = O(W^* 2^k)$ operations, since K_0 is a constant.

Recursive step. If $k \geq K_0$, write $k = mf + r$ with $f = \lfloor k/m \rfloor \geq a+1$ and $0 \leq r < m$. Apply the r leftover factors of C directly, then execute the certificate on f columns with r spectator bits. Gates and translations are pointwise and cost $O(W^* 2^k)$ in total. An edge move $(r, \tau, z_1, \ldots, z_\rho)$ is executed by Lemma 3.1 and Lemma 3.3:

- permute the addresses of role r;
- apply $C^{\otimes \rho f}$ to each of the $2^{k-\rho f}$ contiguous fibres;
- permute back and translate by τ.

Since $\rho \leq m-1$, $k - \rho f \geq f \geq a$, so the fibres split into exactly $2^{k-\rho f-a}$ groups of W^*. Each group is transformed by one recursive call of $\mathcal{A}$ at size $\rho f < k$. Correctness follows from the certificate identity for f columns, which holds for arbitrary contents of every role, and from the correctness of the recursive calls.

Cost. Everything except the recursive calls costs at most $A \cdot W^* 2^k$ for a constant A, as in [1, Theorem 2.6]. With $N(c)$ the number of edge moves of size c and $T(k)$ the cost of $\mathcal{A}$, we get $T(k) \leq AW^*2^k + \sum_c N(c) 2^{k-cf-a} T(cf)$. By induction $T(k) \leq A' W^* G(k)$ with G as in Definition 3.5, and Lemma 3.6 gives $T(k) = O(W^* 2^k (k+1)^z)$.

Depth and workspace. Along every recursion path $k' = \rho\lfloor k/m \rfloor \leq k - \lfloor k/m \rfloor \leq k-1$. So consecutive array lengths at least halve, and the workspace is $O(W^* 2^k)$ as in [1]. The depth is $O(\log(k/K_0)/\log \frac{m}{m-1}) = O(\log k)$, so stacks and counters fit in $O(\log n)$-bit words. □

Corollary 3.8. *Under the hypotheses of Theorem 3.7 with $0 < z \leq 1$, F_n and exact convolution cost $O\big(n(\log n)^z (\log\log n)^{4-z}\big)$, and hence $O(n(\log n)^q)$ for every $q > z$.*

Proof. Proposition 2.1 and Corollary 2.2. □

Lemma 3.9 (grouping never hurts)**.** *Let $\mathcal{E}$ be as in Theorem 3.7, with $s = \sum_\rho \rho < W^* m$ and $|\mathcal{E}| > W^*$. Then $F(z) = \sum_\rho (\rho/m)^z$ is continuous and strictly decreasing on $[0,1]$, with $F(0) > W^* > F(1)$. Its unique root p of $F(p) = W^*$ satisfies $p \leq \theta = \log_m(s/W^*)$.*

Proof. Each $0 < \rho/m < 1$. We have $F(0) = |\mathcal{E}| > W^*$ and $F(1) = s/m < W^*$. For $\rho \geq 1$, $\rho^\theta \leq \rho$, so $F(\theta) \leq m^{-\theta} s = W^* = F(p)$, and monotonicity gives $p \leq \theta$. □

Remark 3.10 (size of the gain). For $z = 1 - \varepsilon$ close to 1, $(\rho/m)^z \approx (\rho/m)(1 + \varepsilon \ln(m/\rho))$. Writing $\Delta = W^* m - s$ for the saving,

$$1 - \theta \approx \frac{\Delta}{W^* m \ln m}, \qquad 1 - p \approx \frac{\Delta/m}{\sum_\rho \frac{\rho}{m} \ln \frac{m}{\rho}}.$$

So $(1-p)/(1-\theta)$ is about $\ln m$ divided by the ρ-weighted mean of $\ln(m/\rho)$. It is large when most of s sits in edges of size close to m.

3.4 Certifying the rate inequality

Lemma 3.11 (exponent certificate)**.** *Let $m \geq 2$, $L \geq \ln m$, $0 < \varepsilon < 1/L$ and $\mu = 1/(1-\varepsilon L)$. For each $1 \leq c \leq m$ let $\ell_c \geq \ln(m/c)$. Then*

$$\left(\frac{c}{m}\right)^{1-\varepsilon} \leq \frac{c}{m}\bigl(1+\varepsilon\mu\ell_c\bigr).$$

Consequently (2) holds for $z = 1-\varepsilon$ whenever $\sum_\rho \frac{\rho}{m}(1+\varepsilon\mu\ell_\rho) < W^*$, a finite inequality between rationals when ε, L and the ℓ_ρ are rational.

Proof. Put $y = \varepsilon\ln(m/c) \in [0, \varepsilon L]$. Then $(c/m)^{-\varepsilon} = e^y \leq 1/(1-y) = 1+y/(1-y) \leq 1+y/(1-\varepsilon L) \leq 1+\varepsilon\mu\ell_c$. □

Remark 3.12. Upper bounds ℓ_c are obtained from $\ln 2 < 0.6931471808$ together with $\ln y \leq y-1$, or from partial sums of the series of ln.

4 OpenAI's formalized network

OpenAI's formalization [2] does not use the network of [1, Section 2]. It uses a smaller explicit one:

- three factors of 32 binary coordinates each, so $m = 32^3 = 32768$;
- the $v = \binom{30}{3} = 4060$ triples on 30 of the coordinates of each factor;
- 33 centres;
- one side wire for every ordered pair of triples;
- $W^* = 2^{50}$ roles.

Its certificate has $2^{50}m - \Delta$ directional moves with $\Delta = 29\,406\,742\,400$, and the formalization proves the per-direction exponent $1 - 2\cdot 10^{-11}$ (the true value of $1-\theta$ is $7.666\cdot 10^{-11}$).

Proposition 4.1 (grouped certificate)**.** *OpenAI's formalized network yields a $(32768, 2^{50})$-certificate whose edge multiset is given by Table 1. All sizes lie in $[1, m-1]$, and $\sum_\rho \rho = 2^{50}m - \Delta$ with the same $\Delta = 29\,406\,742\,400$.*

Proof. Every label of the formalized network is a subset of a fixed orthonormal basis of its role, and consecutive labels of a role differ by adding or removing a set of basis lines. Grouping each such change into one edge move (Lemma 3.1 with M as in the remark after it) gives a certificate. Gates, scalar identities and terminal corrections are those of [2]. There are two places where a role passes from the empty label to the full one in one step: the unused side wires at the end of a stage, and the padding roles. There the change is routed through an intermediate label of dimension 1, giving sizes 1 and $m-1$. The multiplicities follow from the cardinalities of the labels of [2]. The bookkeeping is routine; it is part of the formal verification (Section 7). The value of $\sum\rho$ follows from the table. □

Proposition 4.2 (rate)**.** *For $\varepsilon = 1.4657\cdot 10^{-8}$, $\sum_c N(c)(c/m)^{1-\varepsilon} < 2^{50}$.*

Proof. Apply Lemma 3.11 with $L = 16 > \ln m = 15\ln 2$ and $\ell_c = j_c\cdot 0.6931471808 + (m/(2^{j_c}c) - 1)$, where $2^{j_c} \leq m/c < 2^{j_c+1}$. The resulting rational inequality has 13 terms and is verified exactly; this is part of the formal verification (Section 7). The exact root of $F(p) = 2^{50}$ is $1-p = 1.4710\cdot 10^{-8}$. □

size c	multiplicity $N(c)$	value
1	$4 \cdot 4060^2\, n_b + 3 \cdot 4060^2\, n_u + (2^{50} - W)$	$1.3269 \cdot 10^{15}$
30	$3 \cdot 4060^2\, n_b$	$6.0352 \cdot 10^{14}$
31	$6 \cdot 4060^3 + 4060^2\, n_b$	$2.0157 \cdot 10^{14}$
32	$9 \cdot 33 \cdot 4060^2$	$4.8956 \cdot 10^{9}$
961	$2 \cdot 4060^3$	$1.3385 \cdot 10^{11}$
962	$4060^2\, n_b$	$2.0117 \cdot 10^{14}$
992	$33 \cdot 4060^2$	$5.4396 \cdot 10^{8}$
1023	$4060^2\, n_b$	$2.0117 \cdot 10^{14}$
31713	$2 \cdot 4060^3$	$1.3385 \cdot 10^{11}$
31714	$4060^2\, n_b$	$2.0117 \cdot 10^{14}$
31744	$4060^2\, n_b + 33 \cdot 4060^2$	$2.0117 \cdot 10^{14}$
32736	$4060^2\, n_b + 66 \cdot 4060^2$	$2.0117 \cdot 10^{14}$
32767	$3 \cdot 4060^2\, n_u + (2^{50} - W)$	$5.2225 \cdot 10^{14}$

Table 1: Edge multiset of OpenAI's formalized network under grouping. Notation: 4060^2 is the number of invocations per stage; $n_b = 4060 \cdot 3006$ and $n_u = 4060 \cdot 1054$ are the numbers of used (neighbouring) and unused side pairs per invocation; 33 is the number of centres; $W = 2 \cdot 4060^3 + 3 \cdot 4060^2(4060^2 + 33)$ is the number of network roles. The $2^{50} - W$ padding roles contribute one edge of size 1 and one of size $m - 1$ each. In total $|\mathcal{E}| = 3.66 \cdot 10^{15}$.

Proof of Theorem 1.2. By Propositions 4.1 and 4.2, Theorem 3.7 applies with $z = 1 - 1.4657 \cdot 10^{-8}$. Corollary 3.8 with $q = 1 - 10^{-8} > z$ gives the bound. The formal proof follows exactly this route (Section 7). □

5 Colour registers

In the network of [1, §2], every invocation carries one side wire A_{ST} for each ordered pair of neighbouring triples. That is $d = \binom{h-3}{3} + 3(h-3)$ auxiliary wires per bank element, and these wires make up almost all of the role count W. In this section we replace most of them by *registers*. Each bank element owns a few auxiliary wires. The registers of many source elements meet the registers of many target elements at a common binary label, where a single gate can apply an arbitrary transfer matrix between them. A twelve-step schedule realizes the required transfer for arbitrary auxiliary contents (Lemma 5.13), and the stage-2 invocation is its inverse-transpose (Lemma 5.14). For the stage sizes $(h_1, h_2, h_3) = (30, 20, 41)$ the network uses about 57, 40 and 73 auxiliary wires per bank element, instead of 3006, 731 and 8550. It satisfies the hypotheses of the grouped tensor-power theorem with exponent $z = 1 - 3.21 \cdot 10^{-6}$ (Theorem 5.30).

We use the notation of [1, §2] throughout and cite its results without re-proving them. The construction is in Sections 5.1–5.6: factor data and centres, meeting labels, register designs, the invocations, labels and edges, and the network. The designs follow in Section 5.7, and the exponent and the main theorem in Sections 5.8–5.9. Each statement below is proved here, cited, or explicitly marked as verified by exhaustive computation. The whole argument of this section is part of the formal verification described in Section 7.

5.1 Factor data and centres

Fix integers $h_1, h_2, h_3 \geq 9$, indexed by the stage order. For $j = 1, 2, 3$ let $D_j = \mathbb{F}_2^{h_j}$ with its ordinary dot product, and let

$$\mathbb{D} = D_1 \otimes D_2 \otimes D_3 = \mathbb{F}_2^m, \qquad m = h_1 h_2 h_3,$$

with the tensor dot product, which is the ordinary dot product of $\mathbb{F}_2^m$ [1, §2.3]. For a binary vector z we have $z \cdot z \equiv \mathrm{wt}(z) \pmod 2$, so z has norm one if and only if it has odd weight. A subspace is *nondegenerate* if the dot product restricted to it has zero radical. An *orthonormal basis* (ONB) is a basis $z_1, \ldots, z_r$ with $z_i \cdot z_k = \delta_{ik}$. Orthogonal complements are taken in the ambient space indicated (D_j, a tensor factor, or $\mathbb{D}$). For nested subspaces $U \subseteq V$, the *residual* is $V \cap U^\perp$ [1, Lemma 2.2]. We use two standard facts about symmetric bilinear forms over $\mathbb{F}_2$.

(F1) If $U \subseteq V$ and U is nondegenerate, then $V = U \perp (V \cap U^\perp)$ (an orthogonal direct sum), so $\dim(V \cap U^\perp) = \dim V - \dim U$. The residual $V \cap U^\perp$ is nondegenerate if and only if V is. If $V = U \perp W$ is any orthogonal decomposition with U nondegenerate, then $V \cap U^\perp = W$. In a nondegenerate ambient space of dimension n, $\dim U^\perp = n - \dim U$.

(F2) The Gram matrix of a tensor product of bases is the Kronecker product of the Gram matrices. Hence tensor products of nondegenerate subspaces are nondegenerate, and tensor products of ONBs are ONBs. An orthogonal sum is nondegenerate if and only if its summands are, and the union of ONBs of the summands is an ONB of the sum.

Definition 5.1 (factor data)**.** Fix j and write $h = h_j$, $D = D_j$. Let $\mathcal{T} = \mathcal{T}_j$ be the set of 3-subsets of $[h] = \{1, \ldots, h\}$, $v = v_j = \binom{h}{3}$, and $t_T \in D$ the indicator of $T \in \mathcal{T}$, so $t_T \cdot t_T = 1$. As in [1, §2.1], $S, T \in \mathcal{T}$ are *neighbours* if $|S \cap T|$ is even, equivalently $t_S \cdot t_T = 0$; neighbours are distinct. An *ordered orthogonal pair* is an ordered pair (T, S) of neighbours, so $|T \cap S| \in \{0, 2\}$. Let $\Omega = \Omega_j$ be the set of ordered orthogonal pairs, Ω^0 the subset with $T \cap S = \emptyset$, and Ω^2 the subset with $|T \cap S| = 2$. Then $\Omega = \Omega^0 \sqcup \Omega^2$, with $|\Omega^0| = v\binom{h-3}{3}$ and $|\Omega^2| = 3v(h-3)$.

The banks are those of [1, §2.1]: wires X_d, Y_d for $d \in \mathcal{T}_1 \times \mathcal{T}_2 \times \mathcal{T}_3$, with $N_0 = v_1 v_2 v_3$, $u_d = t_{d_1} \otimes t_{d_2} \otimes t_{d_3}$ and $U_d = \langle u_d \rangle$. At stage j, an *invocation* is indexed by $o \in \mathcal{O}_j := \prod_{l \neq j} \mathcal{T}_l$, so there are N_0 / v_j invocations. For $T \in \mathcal{T}_j$ let (o, T) be the key with j-th entry T and other entries o. The invocation o has *source bank* $x_T := X_{(o,T)}$ and *target bank* $y_S := Y_{(o,S)}$ $(T, S \in \mathcal{T}_j)$. Its stage spaces are those of [1, §2.3]:

$$E = \bigotimes_{l<j} D_l, \quad P = \langle p \rangle,\ p = \bigotimes_{l<j} t_{o_l}, \quad B = P^\perp \cap E, \quad F = \bigotimes_{l>j} D_l, \quad Q = \langle q \rangle,\ q = \bigotimes_{l>j} t_{o_l}.$$

An empty tensor product is $\mathbb{F}_2$. Thus $p \cdot p = q \cdot q = 1$ and $E = B \perp P$, with $B = 0$ at stage 1 and $Q = F = \mathbb{F}_2$ at stage 3. Put $\alpha_j = \dim E = \prod_{l<j} h_l$ and $\beta_j = \dim F = \prod_{l>j} h_l$. Then $m = \alpha_j h_j \beta_j$ and $\dim B = \alpha_j - 1$.

Definition 5.2 (centres)**.** With h, $\mathcal{T}$ as above, let $G \colon \mathbb{C}^{\mathcal{T}} \to \mathbb{C}^h$ and $R \colon \mathbb{C}^h \to \mathbb{C}^{\mathcal{T}}$ be

$$(Gx)_i = \tfrac{1}{2}\Big(\sum_{T \ni i} x_T - \tfrac{1}{3} \sum_{T \in \mathcal{T}} x_T \Big), \qquad (Rc)_S = \sum_{i \in S} c_i,$$

and for $S, T \in \mathcal{T}$ put $K_{ST} := \frac{1}{2}\big(1 - |S \cap T|\big)$. Thus $K_{ST} = K_{TS}$; it equals $\frac{1}{2}$ on disjoint pairs and $-\frac{1}{2}$ on pairs meeting in two points.

Lemma 5.3 (h centres)**.** *For all $S, T \in \mathcal{T}$,*

$$(RG)_{ST} = \tfrac{1}{2}\big(|S \cap T| - 1\big) = -K_{ST}.$$

Consequently $RG + K \circ 1_\Omega = I$, where $(K \circ 1_\Omega)_{ST} = K_{ST}$ if $(T, S) \in \Omega$ and 0 otherwise.

Proof. Since $|S| = 3$, $(RG)_{ST} = \sum_{i \in S} G_{iT} = \frac{1}{2} \sum_{i \in S} \left([i \in T] - \frac{1}{3} \right) = \frac{1}{2}(|S \cap T| - 1)$. Consider the entry (S, T) of $RG + K \circ 1_\Omega$. If $S = T$, then $|S \cap T| = 3$, so $(RG)_{TT} = 1$, and $(T, T) \notin \Omega$. If $|S \cap T| = 1$, then $(RG)_{ST} = 0$ and $(T, S) \notin \Omega$. If $|S \cap T| \in \{0, 2\}$, then $(T, S) \in \Omega$ and the entry is $(RG)_{ST} + K_{ST} = 0$. □

Remark 5.4. OpenAI's network uses $h + 1$ centres $c_1, \ldots, c_h, c_*$ and obtains the same matrix $(RG)_{ST} = (|S \cap T| - 1)/2$ [1, §2.1 and proof of Lemma 2.1]. Lemma 5.3 shows that c_* is unnecessary: each invocation needs only h_j centres. OpenAI's side-wire map is $(JA)_S = -\sum_T \frac{|S \cap T| - 1}{2} A_{ST} = \sum_T K_{ST} A_{ST}$, the sum over the neighbours T of S.

5.2 Meeting labels

Fix a stage j and an invocation o, and write $D = D_j$, $h = h_j$, $\alpha = \alpha_j$, $\beta = \beta_j$.

Definition 5.5 (meeting labels)**.** For a subspace $L \subseteq D$ put

$$\Lambda_L := \big((B \otimes D) \perp (P \otimes L) \big) \otimes Q \ \subseteq \ \mathbb{D}.$$

The special labels are

$$\mathrm{Bot} := \Lambda_0 = B \otimes D \otimes Q, \qquad \mathrm{Xb}(t) := \Lambda_{\langle t \rangle}, \qquad \mathrm{Yt}(t) := \Lambda_{t^\perp}, \qquad \mathrm{Top} := \Lambda_D = E \otimes D \otimes Q,$$

for $t = t_T$, with $t^\perp$ taken in D.

These are labels of [1, (2.7)], with D replaced by D_j. Bot is the row-1 and row-4 label. $\mathrm{Xb}(t)$ is the row-2 label. $\mathrm{Yt}(t)$ is the row-5 label and the outgoing label of the target bank. Top is the row-3, row-6 and row-7 label and the outgoing label of the source bank. The row-0 label $B \otimes \langle t \rangle \otimes Q$ of [1] is replaced by Bot.

Lemma 5.6 (meeting labels)**.** *Let $L \subseteq L' \subseteq D$ be subspaces.*

(i) $\Lambda_L = \mathrm{Bot} \perp (P \otimes L \otimes Q)$*, and $z \mapsto p \otimes z \otimes q$ is an isometric linear embedding of D onto $P \otimes D \otimes Q$. Hence Λ_L is nondegenerate if and only if L is, and $\Lambda_L \subseteq \Lambda_{L'}$. In particular* Bot*,* $\mathrm{Xb}(t)$*,* $\mathrm{Yt}(t)$ *and* Top *are nondegenerate.*

(ii) If L and L' are nondegenerate, the residual of $\Lambda_L \subseteq \Lambda_{L'}$ is $P \otimes (L' \cap L^\perp) \otimes Q$, of dimension $\dim L' - \dim L$*.*

(iii) If $z_1, \ldots, z_r$ is an ONB of $N \subseteq D$, then $p \otimes z_1 \otimes q, \ldots, p \otimes z_r \otimes q$ is an ONB of $P \otimes N \otimes Q$.

(iv) [1, end of §2.3] A nondegenerate subspace $N \subseteq \mathbb{F}_2^n$ has an ONB if and only if $N = 0$ or N contains a vector of odd weight.

Proof. (i) $B \perp P$ gives $B \otimes D \perp P \otimes L$. Moreover $(p \otimes z \otimes q) \cdot (p \otimes z' \otimes q) = (p \cdot p)(z \cdot z')(q \cdot q) = z \cdot z'$, and the map is injective because $p, q \neq 0$. So $P \otimes L \otimes Q$ is isometric to L. The space B is nondegenerate by (F1), being the residual of the nondegenerate line P in E. Hence Bot is nondegenerate by (F2), and Λ_L is nondegenerate if and only if L is, again by (F2). The subspaces 0, $\langle t \rangle$, $t^\perp$ and D are nondegenerate, $t^\perp$ by (F1). Monotonicity is clear. (ii) By (F1), $L' = L \perp N$ with $N = L' \cap L^\perp$. Applying the isometry of (i), $\Lambda_{L'} = \Lambda_L \perp (P \otimes N \otimes Q)$, and the last part of (F1) identifies the residual. Its dimension is $\dim N = \dim L' - \dim L$. (iii) follows from the isometry in (i). (iv) is the splitting statement at the end of [1, §2.3], with "norm one" read as "odd weight". The converse is trivial, since ONB vectors have norm one. □

5.3 Register designs

Definition 5.7 (register design)**.** Let $h \geq 9$, $D = \mathbb{F}_2^h$, and let $\mathcal{T}$, Ω be as in Definition 5.1. A *register design* $\mathcal{R}$ for D consists of finite sets $\mathcal{X}$ (X-registers), $\mathcal{Y}$ (Y-registers), Tr (transfers) and Σ (side pairs) with the following data.

- Each $\xi \in \mathcal{X}$ has an *owner* $T_\xi \in \mathcal{T}$ and a *chain* $L_{\xi,1} \subsetneq \cdots \subsetneq L_{\xi,k_\xi}$ of subspaces of D, with $k_\xi \geq 1$.
- Each $\eta \in \mathcal{Y}$ has an owner $S_\eta \in \mathcal{T}$ and a chain $L_{\eta,1} \subsetneq \cdots \subsetneq L_{\eta,k_\eta}$, with $k_\eta \geq 1$.
- Each $\tau \in \mathrm{Tr}$ has an X-register ξ_τ, a Y-register η_τ and a label $L_\tau \subseteq D$. Its *owner pair* is $(T_{\xi_\tau}, S_{\eta_\tau})$.
- Each $\sigma \in \Sigma$ is an ordered orthogonal pair $(T_\sigma, S_\sigma) \in \Omega$.

The *meeting labels* are the subspaces that occur in chains. For $T, S \in \mathcal{T}$ the *cover count* is

$$n(T,S) = \#\{\tau : (T_{\xi_\tau}, S_{\eta_\tau}) = (T,S)\} + \#\{\sigma : (T_\sigma, S_\sigma) = (T,S)\}.$$

The design must satisfy:

(D1) every meeting label is nondegenerate;

(D2) for every X-register with owner T and chain $L_1 \subsetneq \cdots \subsetneq L_k$: $t_T \in L_1$, and each of the spaces $L_1 \cap t_T^\perp$, $L_{i+1} \cap L_i^\perp$ $(1 \leq i < k)$ and $L_k^\perp$ is 0 or contains a vector of odd weight;

(D3) for every Y-register with owner S and chain $L_1 \subsetneq \cdots \subsetneq L_k$: $L_k \subseteq t_S^\perp$, and each of the spaces L_1, $L_{i+1} \cap L_i^\perp$ $(1 \leq i < k)$ and $t_S^\perp \cap L_k^\perp$ is 0 or contains a vector of odd weight;

(D4) for every transfer τ, the label L_τ belongs to the chain of ξ_τ and to the chain of η_τ;

(D5) $n(T,S) = 1$ for $(T,S) \in \Omega$, and $n(T,S) = 0$ for $(T,S) \notin \Omega$.

For $\Omega' \subseteq \Omega$, a *partial register design for* Ω' satisfies (D1)–(D4) and (D5$_{\Omega'}$): $n(T,S) = 1$ for $(T,S) \in \Omega'$ and $n(T,S) = 0$ otherwise. We write $\mathrm{regs}(\mathcal{R}) = |\mathcal{X}| + |\mathcal{Y}|$ and $\mathrm{side}(\mathcal{R}) = |\Sigma|$.

If $\mathcal{X} = \mathcal{Y} = \mathrm{Tr} = \emptyset$ and $\Sigma = \Omega$, (D1)–(D5) hold trivially. This design gives OpenAI's side wires, one for each ordered orthogonal pair. The point of registers is that one meeting label can serve many pairs: if w X-registers and r Y-registers share a meeting label, transfers at that label can cover up to wr ordered pairs with $w + r$ wires.

Lemma 5.8 (union)**.** *If $\mathcal{R}'$ and $\mathcal{R}''$ are partial register designs for disjoint sets Ω', Ω'', then their disjoint union (of registers, transfers and side pairs) is a partial register design for $\Omega' \cup \Omega''$. In particular, a partial design for Ω^0 together with one for Ω^2 is a register design.*

Proof. Conditions (D1)–(D4) concern individual registers and transfers, and the cover counts add. □

Lemma 5.9 (first-cover designs)**.** *Let $\Omega' \subseteq \Omega$, and let $\mathcal{X}$, $\mathcal{Y}$ be finite sets of registers (owners and chains) satisfying (D1)–(D3). Let $\mathcal{C}$ be a finite, totally ordered set of candidates $\gamma = (L_\gamma, \xi_\gamma, \eta_\gamma)$, where L_γ lies in the chains of both ξ_γ and η_γ and the owner pair $(T_{\xi_\gamma}, S_{\eta_\gamma})$ lies in Ω'. Call a pair covered if it is the owner pair of some candidate. Define:*

- Tr*: for each covered pair, the first candidate with that owner pair;*

- Σ*: the pairs of* Ω' *that are not covered.*

Then $(\mathcal{X}, \mathcal{Y}, \mathrm{Tr}, \Sigma)$ *is a partial register design for* Ω'*, and* $|\Sigma| = |\Omega'| - \#\{\textit{covered pairs}\}$.

Proof. (D1)–(D3) are assumed, and (D4) holds because every candidate's label lies in both chains. A pair of Ω' is covered by exactly one transfer if it is covered, and by exactly one side pair otherwise. A pair outside Ω' is the owner pair of no candidate and is not in Σ. □

Remark 5.10 (first-cover semantics). In Lemma 5.9, a candidate that is not selected still has both registers passing through its label. In step 5 of the schedule below the two registers then meet at that label with coefficient 0, which is harmless. A design can therefore be specified by its registers and an ordered candidate list. The transfer of a pair is assigned to its first covering candidate, and the side pairs are *defined* as the uncovered pairs. With these semantics, the exact cover (D5) holds by construction. The only global quantity that must be computed is the number $|\Sigma|$ of uncovered pairs, which enters the role count.

5.4 The invocations

Fix a stage j, an invocation o, and data $(\mathcal{X}, \mathcal{Y}, \mathrm{Tr}, \Sigma)$ as in Definition 5.7 for $D = D_j$. Conditions (D1)–(D5) are imposed only where stated. The wires of the invocation are:

- the source and target bank entries x_T, y_S $(T, S \in \mathcal{T})$;
- centres $c_1, \dots, c_h$;
- one side wire A_σ for each $\sigma \in \Sigma$;
- one X-register wire a_ξ for each $\xi \in \mathcal{X}$ and one Y-register wire b_η for each $\eta \in \mathcal{Y}$.

All but the bank entries are private to the invocation. Besides G and R we use the linear maps

$$(\Pi x)_\xi = x_{T_\xi}, \qquad (Vx)_\sigma = x_{T_\sigma}, \qquad (\Xi b)_S = \sum_{\eta:\, S_\eta = S} b_\eta, \qquad (JA)_S = \sum_{\sigma:\, S_\sigma = S} K_{ST_\sigma} A_\sigma,$$

$$(\widetilde{K}_L a)_\eta = \sum_{\tau:\, \eta_\tau = \eta,\ L_\tau = L} K_{S_\eta T_{\xi_\tau}}\, a_{\xi_\tau}, \qquad \widetilde{K} = \sum_L \widetilde{K}_L.$$

So Π copies x_T into the X-registers of T, Ξ sums the Y-registers of S into y_S, and $\widetilde{K}$ is the *transfer matrix*: the entry K_{ST} goes from the X-register to the Y-register of each transfer with owner pair (T, S). When $\Sigma = \Omega$, the map J is OpenAI's (Remark 5.4).

Definition 5.11 (forward invocation)**.** The *forward invocation* is the chronological sequence of gates in Table 2. A gate is a linear map $I + Z$ of the vector of wire values. Z is the displayed increment, and it is supported on the listed wires, which the gate *touches*. The gate's label is the listed subspace. In step 5 there is one gate for every meeting label L, with increment $\widetilde{K}_L$. It touches all registers whose chain contains L, and the gates are applied in order of nondecreasing $\dim L$, with ties in any fixed order. Under (D4), $\widetilde{K}_L$ acts only on registers whose chains contain L, so its support lies in the touched wires.

Gates of one step touch pairwise disjoint sets of wires, except in step 5. There, two gates at distinct meeting labels of equal dimension touch disjoint sets of registers, because a strictly increasing chain contains at most one subspace of each dimension. Every increment Z maps wires of one kind into wires of a different kind, as in $x \to c$, $c \to y$, $\{b, A\} \to y$, $x \to \{a, A\}$ and $a \to b$. Hence $Z^2 = 0$ for every gate.

step	update	gates and touched wires	label
1	$y \leftarrow y - Rc$	one gate: all y_S and all c_i	Bot
2	$y \leftarrow y - \Xi b - JA$	one gate per S: y_S; b_η with $S_\eta = S$; A_σ with $S_\sigma = S$	Bot
3	$b \leftarrow b - \widetilde{K}a$	one gate: all registers	Bot
4	$a \leftarrow a + \Pi x$, $A \leftarrow A + Vx$	one gate per T: x_T; a_ξ with $T_\xi = T$; A_σ with $T_\sigma = T$	$\mathrm{Xb}(t_T)$
5	$b \leftarrow b + \widetilde{K}_L a$	one gate per meeting label L, by nondecreasing $\dim L$: all registers whose chain contains L	Λ_L
6	$c \leftarrow c + Gx$	one gate: all x_T and all c_i	Top
7	$y \leftarrow y + Rc$	one gate: all y_S and all c_i	Bot
8	$y \leftarrow y + \Xi b + JA$	one gate per S, touching the same wires as in step 2	$\mathrm{Yt}(t_S)$
9	$c \leftarrow c - Gx$	as in step 6	Top
10	$b \leftarrow b - \widetilde{K}a$	as in step 3	Top
11	$a \leftarrow a - \Pi x$, $A \leftarrow A - Vx$	one gate per T, touching the same wires as in step 4	Top
12	$b \leftarrow b + \widetilde{K}a$	as in step 3	Top

Table 2: The forward invocation (stages 1 and 3). All labels include the factor Q (Definition 5.5).

Remark 5.12. If $\mathcal{X} = \mathcal{Y} = \emptyset$, steps 3, 5, 10 and 12 are empty. The remaining steps are OpenAI's rows [1, (2.3)] in the order $1, 0, 2, 3, 4, 5, 6, 7$, with h centres and the row-0 label Bot. Rows 0 and 1 commute, since both only subtract from y, using wires that neither modifies.

Lemma 5.13 (scalar identity)**.** *For arbitrary initial values of all wires, the forward invocation maps*

$$(x, y, c, A, a, b) \longmapsto (x,\ y + Mx,\ c, A, a, b), \qquad M = RG + JV + \Xi\widetilde{K}\Pi,$$

where $M_{ST} = (RG)_{ST} + K_{ST}\, n(T, S)$. If (D5) holds, then $M = I$. That is, the forward invocation maps $(x, y) \mapsto (x, y + x)$ and restores every auxiliary wire.

Proof. No gate writes x. Let $(x, y_0, c_0, A_0, a_0, b_0)$ be the initial values. In step 5 every gate writes only b and reads only a, and a is not written in step 5. So step 5 as a whole adds $\sum_L \widetilde{K}_L a = \widetilde{K}a$ to b, whatever the order of its gates. Steps with several gates act on disjoint wires, so each step performs the displayed update. Table 3 records the wires written in each step and their values afterwards. Every entry follows from the previous ones and the update of the step.

Now compute M. By definition, $(JVx)_S = \sum_{\sigma:\, S_\sigma = S} K_{ST_\sigma} x_{T_\sigma}$, so $(JV)_{ST} = K_{ST} \cdot \#\{\sigma : (T_\sigma, S_\sigma) = (T, S)\}$. Also $(\Xi\widetilde{K}\Pi x)_S = \sum_{\eta:\, S_\eta = S} \sum_{\tau:\, \eta_\tau = \eta} K_{ST_{\xi_\tau}} x_{T_{\xi_\tau}}$, so $(\Xi\widetilde{K}\Pi)_{ST} = K_{ST} \cdot \#\{\tau : (T_{\xi_\tau}, S_{\eta_\tau}) = (T, S)\}$. Hence $M_{ST} = (RG)_{ST} + K_{ST}\, n(T, S)$. Under (D5), $n(T, S) = [(T, S) \in \Omega]$, and Lemma 5.3 gives $M = I$. □

Lemma 5.14 (reverse invocation)**.** *Define the reverse invocation by replacing each gate $I + Z$ of the forward invocation by $I - Z^{\mathsf{T}}$, keeping its touched wires, its label and its place in the chronological order. Assume (D5). For arbitrary initial values of all wires the reverse invocation maps*

$$(x, y, c, A, a, b) \longmapsto (x - y,\ y,\ c, A, a, b).$$

Every wire has the same sequence of touches and labels in both invocations, so the two invocations have the same edges and residuals.

after step	wires written	value
1	y	$y_0 - Rc_0$
2	y	$y_0 - Rc_0 - \Xi b_0 - JA_0$
3	b	$b_0 - \widetilde{K} a_0$
4	a, A	$a_0 + \Pi x, \quad A_0 + Vx$
5	b	$b_0 - \widetilde{K} a_0 + \widetilde{K}(a_0 + \Pi x) = b_0 + \widetilde{K}\Pi x$
6	c	$c_0 + Gx$
7	y	$y_0 - Rc_0 - \Xi b_0 - JA_0 + R(c_0 + Gx) = y_0 + RGx - \Xi b_0 - JA_0$
8	y	$y_0 + RGx - \Xi b_0 - JA_0 + \Xi(b_0 + \widetilde{K}\Pi x) + J(A_0 + Vx) = y_0 + Mx$
9	c	c_0
10	b	$b_0 + \widetilde{K}\Pi x - \widetilde{K}(a_0 + \Pi x) = b_0 - \widetilde{K} a_0$
11	a, A	$a_0, \quad A_0$
12	b	$b_0 - \widetilde{K} a_0 + \widetilde{K} a_0 = b_0$

Table 3: Values in the proof of Lemma 5.13.

Proof. Let $N_1, \dots, N_r$ be the gates of the forward invocation in chronological order. By Lemma 5.13 and (D5), $N := N_r \cdots N_1 = I + \mathcal{E}$, where $\mathcal{E}$ has entry 1 in position (y_T, x_T) for each T and 0 elsewhere. Since $Z^2 = 0$ for every gate, $(I + Z)^{-1} = I - Z$, so the reverse gates are $N_i^{-\mathsf{T}}$. The reverse invocation is therefore

$$N_r^{-\mathsf{T}} \cdots N_1^{-\mathsf{T}} = \left(N_1^{-1} \cdots N_r^{-1}\right)^{\mathsf{T}} = (N^{-1})^{\mathsf{T}} = (I - \mathcal{E})^{\mathsf{T}} = I - \mathcal{E}^{\mathsf{T}},$$

using $\mathcal{E}^2 = 0$. The map $I - \mathcal{E}^{\mathsf{T}}$ sends $x_T \mapsto x_T - y_T$ and fixes every other wire. The support of Z^{T} lies in the same set of wires as the support of Z, so the touched wires and labels are unchanged by construction. □

Remark 5.15. Explicitly, the reverse invocation reads, for example: step 1 $c \leftarrow c + R^{\mathsf{T}} y$; step 2 $b_\eta \leftarrow b_\eta + y_{S_\eta}$ and $A_\sigma \leftarrow A_\sigma + K_{S_\sigma T_\sigma} y_{S_\sigma}$; step 4 $x_T \leftarrow x_T - \sum_{T_\xi = T} a_\xi - \sum_{T_\sigma = T} A_\sigma$; step 5 $a \leftarrow a - \widetilde{K}_L^{\mathsf{T}} b$; step 6 $x \leftarrow x - G^{\mathsf{T}} c$. Data flow from Y to X, but every X-register still belongs to the X bank and every Y-register to the Y bank, so the physical touch sequence is that of Table 2. This replaces OpenAI's reversal of the rows [1, §2.3, after (2.7)] and makes a separate reverse schedule unnecessary.

5.5 Labels and edges

At stage j the *incoming labels* are $E \otimes \langle t_T \rangle \otimes Q$ on x_T and $B \otimes \langle t_S \rangle \otimes Q$ on y_S. The *outgoing labels* are Top on x_T and $\mathrm{Yt}(t_S)$ on y_S. Each auxiliary wire has source label 0 and sink label $\mathbb{D}$. These are the stage-boundary labels of [1, §2.3], with D replaced by D_j. As shown there, the outgoing labels of stage j are the incoming labels of stage $j + 1$; the incoming labels at stage 1 are U_d (on X_d) and 0 (on Y_d); and the outgoing labels at stage 3 are $\mathbb{D}$ and $U_d^{\perp}$.

Lemma 5.16 (labels and edges)**.** *Assume (D1)–(D4) for the design of factor j. In every invocation at stage j, forward or reverse:*

(a) every gate label is nondegenerate;

(b) the label sequence of every wire is as in Table 4. In particular each register is touched at Bot*, then (X-registers only) at* $\mathrm{Xb}(t_T)$*, then at its chain labels in increasing order, then (Y-registers only) at* $\mathrm{Yt}(t_S)$*, and finally at* Top*;*

(c) consecutive labels along every wire are nested. The nonzero residuals and their dimensions are those of Table 5. Every residual is nondegenerate and has an ONB;

(d) every edge is increasing, except the centre edges from step 6 to step 7, which decrease by $P \otimes D \otimes Q$, of dimension h_j.

wire	label sequence (step numbers in brackets)
x_T	$\text{in} \to \text{Xb}(t_T)\,[4] \to \text{Top}\,[6,9,11] = \text{out}$
y_S	$\text{in} \to \text{Bot}\,[1,2,7] \to \text{Yt}(t_S)\,[8] = \text{out}$
c_i	$0 \to \text{Bot}\,[1] \to \text{Top}\,[6] \to \text{Bot}\,[7] \to \text{Top}\,[9] \to \mathbb{D}$
A_σ	$0 \to \text{Bot}\,[2] \to \text{Xb}(t_{T_\sigma})\,[4] \to \text{Yt}(t_{S_\sigma})\,[8] \to \text{Top}\,[11] \to \mathbb{D}$
a_ξ	$0 \to \text{Bot}\,[3] \to \text{Xb}(t_{T_\xi})\,[4] \to \Lambda_{L_{\xi,1}} \to \cdots \to \Lambda_{L_{\xi,k}}\,[5] \to \text{Top}\,[10,11,12] \to \mathbb{D}$
b_η	$0 \to \text{Bot}\,[2,3] \to \Lambda_{L_{\eta,1}} \to \cdots \to \Lambda_{L_{\eta,k}}\,[5] \to \text{Yt}(t_{S_\eta})\,[8] \to \text{Top}\,[10,12] \to \mathbb{D}$

Table 4: Label sequences in one invocation. "In" and "out" are the stage-boundary labels.

wire	edge	residual	dimension
x_T, $t = t_T$	$\text{in} \to 4$	$B \otimes t^\perp \otimes Q$	$(\alpha-1)(h-1)$
	$4 \to 6$	$P \otimes t^\perp \otimes Q$	$h-1$
y_S, $t = t_S$	$\text{in} \to 1$	$B \otimes t^\perp \otimes Q$	$(\alpha-1)(h-1)$
	$7 \to 8$	$P \otimes t^\perp \otimes Q$	$h-1$
c_i	$\text{source} \to 1$	$B \otimes D \otimes Q$	$(\alpha-1)h$
	$1 \to 6$; $6 \to 7$ (decrease); $7 \to 9$	$P \otimes D \otimes Q$	h each
	$9 \to \text{sink}$	$E \otimes D \otimes Q^\perp$	$\alpha h(\beta-1)$
A_σ, $(T,S) = (T_\sigma, S_\sigma)$	$\text{source} \to 2$	$B \otimes D \otimes Q$	$(\alpha-1)h$
	$2 \to 4$	$P \otimes \langle t_T \rangle \otimes Q$	1
	$4 \to 8$	$P \otimes \langle t_T, t_S \rangle^\perp \otimes Q$	$h-2$
	$8 \to 11$	$P \otimes \langle t_S \rangle \otimes Q$	1
	$11 \to \text{sink}$	$E \otimes D \otimes Q^\perp$	$\alpha h(\beta-1)$
a_ξ, owner T	$\text{source} \to 3$	$B \otimes D \otimes Q$	$(\alpha-1)h$
chain $L_1 \subsetneq \cdots \subsetneq L_k$	$3 \to 4$	$P \otimes \langle t_T \rangle \otimes Q$	1
	$4 \to L_1$	$P \otimes (L_1 \cap t_T^\perp) \otimes Q$	$\dim L_1 - 1$
	$L_i \to L_{i+1}$	$P \otimes (L_{i+1} \cap L_i^\perp) \otimes Q$	$\dim L_{i+1} - \dim L_i$
	$L_k \to 10$	$P \otimes L_k^\perp \otimes Q$	$h - \dim L_k$
	$12 \to \text{sink}$	$E \otimes D \otimes Q^\perp$	$\alpha h(\beta-1)$
b_η, owner S	$\text{source} \to 2$	$B \otimes D \otimes Q$	$(\alpha-1)h$
chain $L_1 \subsetneq \cdots \subsetneq L_k$	$3 \to L_1$	$P \otimes L_1 \otimes Q$	$\dim L_1$
	$L_i \to L_{i+1}$	$P \otimes (L_{i+1} \cap L_i^\perp) \otimes Q$	$\dim L_{i+1} - \dim L_i$
	$L_k \to 8$	$P \otimes (t_S^\perp \cap L_k^\perp) \otimes Q$	$h - 1 - \dim L_k$
	$8 \to 10$	$P \otimes \langle t_S \rangle \otimes Q$	1
	$12 \to \text{sink}$	$E \otimes D \otimes Q^\perp$	$\alpha h(\beta-1)$

Table 5: Residuals of one invocation at stage j ($h = h_j$, $\alpha = \alpha_j$, $\beta = \beta_j$). "L_i" denotes the step-5 touch at Λ_{L_i}, $Q^\perp$ is taken in F, and $\perp$ inside the middle factor is taken in D. All edges not listed have zero residual.

Proof. (a) The gate labels are Bot, Top, Xb(t), Yt(t) and the Λ_L for meeting labels L. They are nondegenerate by Lemma 5.6(i) and (D1).

(b) Read off the touched wires of each step in Table 2. In step 5 the gates are applied in order of nondecreasing dimension, and the dimensions along a chain strictly increase. So each register

meets its chain labels in chain order, and by (D4) each transfer lies in a gate touching both of its registers. By Lemma 5.14 the reverse invocation has the same sequences.

(c) We use the orthogonal decompositions $E = B \perp P$, $D = \langle t \rangle \perp t^\perp$ and $F = Q \perp Q^\perp$, together with (F1) and Lemma 5.6(ii).

- *Banks.* $E \otimes \langle t \rangle \otimes Q = (B \otimes \langle t \rangle \perp P \otimes \langle t \rangle) \otimes Q$ and $\mathrm{Xb}(t) = (B \otimes \langle t \rangle \perp B \otimes t^\perp \perp P \otimes \langle t \rangle) \otimes Q$. Hence the first is contained in the second, with residual $B \otimes t^\perp \otimes Q$. In the same way $B \otimes \langle t \rangle \otimes Q \subseteq \mathrm{Bot}$, with the same residual. The edges $\mathrm{Xb}(t) \subseteq \mathrm{Top}$ and $\mathrm{Bot} \subseteq \mathrm{Yt}(t)$ are the cases $\langle t \rangle \subseteq D$ and $0 \subseteq t^\perp$ of Lemma 5.6(ii).
- *Centres.* $\mathrm{Top} = \mathrm{Bot} \perp P \otimes D \otimes Q$ by Lemma 5.6(i), and $\mathbb{D} = E \otimes D \otimes F = \mathrm{Top} \perp E \otimes D \otimes Q^\perp$.
- *Side wires.* $(T, S) \in \Omega$, so $\langle t_T \rangle \subseteq t_S^\perp$, and Lemma 5.6(ii) applies to $0 \subseteq \langle t_T \rangle \subseteq t_S^\perp \subseteq D$. The residuals are $\langle t_T \rangle$, then $t_S^\perp \cap t_T^\perp = \langle t_T, t_S \rangle^\perp$, then $\langle t_S \rangle$.
- *Registers.* By (D2), $\langle t_T \rangle \subseteq L_1 \subsetneq \cdots \subsetneq L_k \subseteq D$. By (D3), $0 \subseteq L_1 \subsetneq \cdots \subsetneq L_k \subseteq t_S^\perp \subseteq D$. All of these subspaces are nondegenerate by (D1) and (F1). Lemma 5.6(ii) gives the listed residuals and dimensions, using $\dim t_S^\perp = h - 1$.

Every residual is nondegenerate by (F1). It remains to find ONBs. By (F2) and Lemma 5.6(iii), it suffices to find ONBs of the tensor factors.

- D and E have the coordinate ONBs, and $\langle t \rangle$ has the ONB t.
- $t^\perp$ contains e_i for $i \notin T$. Since $(T, S) \in \Omega$, $\langle t_T, t_S \rangle^\perp$ contains e_i for $i \notin T \cup S$; such i exists because $|T \cup S| \leq 6 < h$.
- If $B \neq 0$, it contains a coordinate unit $e_{i_1} \otimes \cdots \otimes e_{i_{j-1}}$ of E with $i_1 \notin o_1$, which is orthogonal to p. Similarly $Q^\perp \neq 0$ contains a coordinate unit of F orthogonal to q [1, proof of Lemma 2.3].
- The register residuals $L_1 \cap t_T^\perp$, $L_{i+1} \cap L_i^\perp$, $L_k^\perp$ (X) and L_1, $L_{i+1} \cap L_i^\perp$, $t_S^\perp \cap L_k^\perp$ (Y) are 0 or contain a vector of odd weight, by (D2) and (D3).

In every case Lemma 5.6(iv) gives an ONB.

(d) Along each sequence of Table 4 the labels increase, except the centres' Top $\to$ Bot between steps 6 and 7. □

Remark 5.17. For banks, centres and side wires, the labels are those of [1, (2.7)], with D_j in place of D, the row-0 label Bot and h centres. If the design has no registers, Lemma 5.16 reduces to [1, Lemma 2.3] for this variant. Conditions (D1)–(D3) can equivalently be expressed by an explicit ONB of D_j adapted to each register's chain. An X-register satisfies (D1) and (D2) if and only if D has an ONB that contains t_T and in which every L_i is spanned by a subset of the basis containing t_T. One direction takes the union of ONBs of the summands of $D = \langle t_T \rangle \perp (L_1 \cap t_T^\perp) \perp (L_2 \cap L_1^\perp) \perp \cdots \perp L_k^\perp$, which exist by (F1) and Lemma 5.6(iv). Conversely, every residual is then spanned by basis vectors. The same holds for Y-registers, with t_S in the basis but outside every L_i.

5.6 The network

Definition 5.18 (colour-register network)**.** Let $h_1, h_2, h_3 \geq 9$ and let $\mathcal{R}_j$ be a register design for D_j $(j = 1, 2, 3)$. The network $\mathcal{N} = \mathcal{N}(\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3)$ has the bank wires X_d, Y_d. For every stage j and every invocation $o \in \mathcal{O}_j$ it also has a private copy of the auxiliary wires of Section 5.4 for $\mathcal{R}_j$: h_j centres, $\mathrm{side}(\mathcal{R}_j)$ side wires and $\mathrm{regs}(\mathcal{R}_j)$ register wires. The stages are run in the order $1, 2, 3$.

Stage j consists of the invocations $o \in \mathcal{O}_j$, with $x_T = X_{(o,T)}$ and $y_S = Y_{(o,S)}$. They are forward invocations at stages 1 and 3 and reverse invocations at stage 2, and their labels are formed from the stage spaces of o. The terminal labels are those of [1, §2.3]. Every source has label 0, except X_d, which has label U_d. Every sink has label $\mathbb{D}$, except Y_d, which has label $U_d^{\perp}$.

The invocations of one stage act on pairwise disjoint sets of wires. Indeed X_d and Y_d belong only to the invocation $o = (d_l)_{l \neq j}$, and the auxiliary wires are private. So the order of the invocations within a stage is immaterial.

Proposition 5.19 (the network)**.** *Let $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3$ be register designs and $\mathcal{N} = \mathcal{N}(\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3)$.*

(a) For arbitrary initial values on all wires, the three stages map $(X_d, Y_d) \mapsto (-Y_d, X_d)$ for every d and restore every auxiliary wire.

(b) Every gate label is nondegenerate, all wires touched by a gate carry its label, and every edge has nested nondegenerate endpoint labels and a residual with an ONB. The only decreasing edges are the centre edges between steps 6 and 7.

(c) The number of wires and the saving $\Delta := Wm - \sum_e \rho_e$ are

$$W = 2N_0 + \sum_{j=1}^{3} \frac{N_0}{v_j}\big(h_j + \mathrm{side}(\mathcal{R}_j) + \mathrm{regs}(\mathcal{R}_j)\big), \qquad \Delta = 2N_0 - 2\sum_{j=1}^{3} \frac{N_0}{v_j} h_j^2,$$

where ρ_e runs over the residual dimensions of all edges.

(d) The multiset of nonzero residual dimensions is the union over j of N_0/v_j copies of the stage-j multiset of Table 6.

Consequently $\mathcal{N}$ satisfies the network interface (2.4), since the proof of [1, Proposition 2.4] applies verbatim. For every $f \geq 1$, $\mathcal{N}$ applies $C^{\otimes mf}$ to each of its W arbitrary arrays, and every edge e is one frame change of residual dimension ρ_e (Lemma 3.1).

class	wires per invocation	nonzero edge dimensions of each wire
banks	$2v_j$	$(\alpha-1)(h-1)$, $h-1$
centres	h_j	$(\alpha-1)h$, h, h, h, $\alpha h(\beta-1)$
side wires	$\mathrm{side}(\mathcal{R}_j)$	$(\alpha-1)h$, 1, $h-2$, 1, $\alpha h(\beta-1)$
X-registers	$\lvert\mathcal{X}_j\rvert$	$(\alpha-1)h$, 1, d_1-1, $d_2-d_1, \ldots, d_k-d_{k-1}$, $h-d_k$, $\alpha h(\beta-1)$
Y-registers	$\lvert\mathcal{Y}_j\rvert$	$(\alpha-1)h$, d_1, $d_2-d_1, \ldots, d_k-d_{k-1}$, $h-1-d_k$, 1, $\alpha h(\beta-1)$

Table 6: Edge multiset of one invocation at stage j ($h = h_j$, $\alpha = \alpha_j$, $\beta = \beta_j$; zero entries omitted). For a register, $d_1 < \cdots < d_k$ are the dimensions of its chain labels. The bank edges are those entering stage j and those inside it.

Proof. (a) By Lemmas 5.13 and 5.14, and because the invocations of a stage act on disjoint wires, stage 1 maps $Y_d \mapsto Y_d + X_d$, stage 2 maps $X_d \mapsto X_d - Y_d$, and stage 3 maps $Y_d \mapsto Y_d + X_d$. Each stage restores every auxiliary wire. Hence $(x, y) \mapsto (x, y + x) \mapsto (-y, y + x) \mapsto (-y, x)$, as in [1, Lemma 2.1].

(b) Within an invocation this is Lemma 5.16. Every auxiliary wire lies in exactly one invocation, where it runs from its source label 0 to its sink label $\mathbb{D}$. A bank wire passes through the three

stages. Its source and sink labels are the incoming labels of stage 1 and the outgoing labels of stage 3, and the outgoing labels of stage j are the incoming labels of stage $j+1$ (Section 5.5). So its edges are exactly the bank edges of Lemma 5.16 for the three stages.

(c) The formula for W counts the wires. For Δ we follow [1, proof of Lemma 2.3]. Along every wire, the signed changes of label dimension telescope to (sink dimension) − (source dimension). The source dimensions sum to N_0 and the sink dimensions to $Wm-N_0$, so the signed sum is $Wm-2N_0$. By (b), the only decreases are one edge of dimension h_j on each of the $(N_0/v_j)h_j$ centres of stage j. Passing to absolute values adds twice these losses: $\sum_e \rho_e = Wm - 2N_0 + 2\sum_j (N_0/v_j)h_j^2$.

(d) Read off Table 5, using $\dim B = \alpha - 1$, $\dim E = \alpha$ and $\dim Q^\perp = \beta - 1$. Each bank wire lies in exactly one invocation per stage.

For the consequence: the proof of [1, Proposition 2.4] uses three things. The first is the complete linear map [1, (2.6)], whose scalar part is the signed bank exchange of (a). The second is the label properties of (b), through [1, Lemma 2.2]. The third is the terminal labels, through the odd weight $\mathrm{wt}(u_d) = 27$. All three hold for $\mathcal{N}$. □

5.7 Designs

We construct partial designs for Ω^2 by a structured construction and for Ω^0 from searched data. Throughout, $h \geq 9$ and $D = \mathbb{F}_2^h$.

Sunflower labels: pairs meeting in two points

Fix an *axis* $\{a, b\} \subseteq [h]$ and put $Q_{ab} = [h] \setminus \{a, b\}$, so $|Q_{ab}| = n := h - 2$. For $x \in Q_{ab}$ let $g_x = e_a + e_b + e_x = t_{\{a,b,x\}}$, and for $P \subseteq Q_{ab}$ let $M(P) = \mathrm{span}\{g_x : x \in P\} \subseteq D$.

Lemma 5.20 (sunflower labels)**.** *Let $P \subseteq Q_{ab}$.*

(a) $g_x \cdot g_y = \delta_{xy}$. So $\{g_x\}_{x\in P}$ is an ONB of $M(P)$, $M(P)$ is nondegenerate of dimension $|P|$, and for $P \subseteq P'$ the residual of $M(P) \subseteq M(P')$ is $M(P' \setminus P)$.

(b) For $x \in P$: $t_{\{a,b,x\}} = g_x \in M(P)$ and $M(P) \cap g_x^\perp = M(P \setminus \{x\})$.

(c) If $P \neq Q_{ab}$, then $M(P)^\perp$ contains e_y for every $y \in Q_{ab} \setminus P$.

(d) For $c \in Q_{ab} \setminus P$: $t_{\{a,b,c\}} \in M(P)^\perp$, and $t_{\{a,b,c\}}^\perp \cap M(P)^\perp = M(P \cup \{c\})^\perp$. This space contains a vector of odd weight if and only if $P \cup \{c\} \neq Q_{ab}$ or h is even. If $P \cup \{c\} = Q_{ab}$ and h is odd, it is nonzero and has no ONB.

Proof. (a) $g_x \cdot g_y \equiv |\{a, b, x\} \cap \{a, b, y\}| \pmod 2$, which is 3 or 2. Orthonormal vectors are independent, so they form an ONB of their span; the residual follows from (F1). (b) follows from (a). (c) For $x \in P$ we have $e_y \cdot g_x = [y = x] = 0$. (d) $g_c \perp M(P)$ by (a), and $g_c^\perp \cap M(P)^\perp = (M(P) + \langle g_c \rangle)^\perp = M(P \cup \{c\})^\perp$. If $P \cup \{c\} \neq Q_{ab}$, use (c). If $P \cup \{c\} = Q_{ab}$, then $M(Q_{ab})^\perp$ has dimension $h - (h-2) = 2$ and contains the independent vectors $w_a = e_a + 1_{Q_{ab}}$ and $w_b = e_b + 1_{Q_{ab}}$. Indeed, for $x \in Q_{ab}$ the only nonzero contributions to $w_a \cdot g_x$ are $e_a \cdot e_a = 1$ and $1_{Q_{ab}} \cdot e_x = 1$, so $w_a \cdot g_x = 0$, and likewise for w_b. So $M(Q_{ab})^\perp = \{0, w_a, w_b, e_a + e_b\}$, with weights $h-1$, $h-1$, 2. It contains an odd vector if and only if h is even, and by Lemma 5.6(iv) it has no ONB for odd h. □

For every axis fix a cyclic order $x_0, \ldots, x_{n-1}$ of Q_{ab}, with indices in $\mathbb{Z}/n$. For $1 \leq \ell \leq n$ let $U(i, \ell) = \{x_i, x_{i+1}, \ldots, x_{i+\ell-1}\}$, the arc of length ℓ starting at x_i. Put

$$\ell_{\max} = n - 1 \ \text{ if } h \text{ is even}, \qquad \ell_{\max} = n - 2 \ \text{ if } h \text{ is odd}.$$

Proposition 5.21 (cyclic sunflower design)**.** *The following data form a partial register design $\mathcal{R}_h^{\mathrm{sun}}$ for Ω^2. For each axis $A = \{a, b\}$ and $i, k \in \mathbb{Z}/n$:*

- *an X-register $\xi_{A,i}$ with owner $A \cup \{x_i\}$ and chain*

$$M(U(i,1)) \subsetneq M(U(i,2)) \subsetneq \cdots \subsetneq M(U(i,\ell_{\max}));$$

- *a Y-register $\eta_{A,k}$ with owner $A \cup \{x_k\}$ and chain*

$$M(U(k-1,1)) \subsetneq M(U(k-2,2)) \subsetneq \cdots \subsetneq M(U(k-\ell_{\max},\ell_{\max}));$$

- *for $i \neq k$ with $\ell := (k-i) \bmod n \leq \ell_{\max}$, a transfer $\tau_{A,i,k} = (M(U(i,\ell)), \xi_{A,i}, \eta_{A,k})$;*
- *if h is odd, a side pair $\sigma_{A,k} = (A \cup \{x_{k+1}\}, A \cup \{x_k\})$.*

It has $3v$ X-registers, $3v$ Y-registers (6 per element), $3v\ell_{\max}$ transfers, and $3v$ side pairs if h is odd (3 per element), none if h is even. Every chain has dimensions $1, 2, \ldots, \ell_{\max}$.

Proof. The arc $U(i,\ell)$ has ℓ elements, and $U(i,\ell+1) = U(i,\ell) \cup \{x_{i+\ell}\}$. The arc $U(k-\ell,\ell) = \{x_{k-\ell}, \ldots, x_{k-1}\}$ does not contain x_k for $\ell \leq n-1$, and $U(k-\ell-1,\ell+1) = U(k-\ell,\ell) \cup \{x_{k-\ell-1}\}$. Both chains are therefore strictly increasing, with dimensions $1, \ldots, \ell_{\max}$ by Lemma 5.20(a).

(D1) holds by Lemma 5.20(a).

(D2) Let $T = A \cup \{x_i\}$, so $t_T = g_{x_i} \in M(U(i,1)) = \langle g_{x_i} \rangle$. The residuals are $M(U(i,1)) \cap t_T^{\perp} = 0$ (part (b)); $M(\{x_{i+\ell}\}) \ni g_{x_{i+\ell}}$, of weight 3 (part (a)); and $M(U(i,\ell_{\max}))^{\perp}$, which contains some e_y by part (c), since $\ell_{\max} < n$.

(D3) Let $S = A \cup \{x_k\}$, so $t_S = g_{x_k}$. Since $x_k \notin U(k-\ell,\ell)$, every chain label lies in $t_S^{\perp}$ by part (a). The residuals $M(\{x_{k-1}\})$ and $M(\{x_{k-\ell-1}\})$ contain vectors of weight 3. By part (d) with $c = x_k$, the last residual is $t_S^{\perp} \cap M(U(k-\ell_{\max},\ell_{\max}))^{\perp} = M(U(k-\ell_{\max},\ell_{\max}+1))^{\perp}$. The arc $U(k-\ell_{\max},\ell_{\max}+1)$ equals Q_{ab} exactly when $\ell_{\max}+1 = n$, that is, when h is even. So part (d) gives an odd vector in both parities.

(D4) If $k \equiv i + \ell$, then $U(i,\ell) = U(k-\ell,\ell)$ is the ℓ-th label of both chains.

(D5$_{\Omega^2}$) Let $(T,S) \in \Omega^2$. Then $A := T \cap S$ is the unique axis contained in both, and $T = A \cup \{x_i\}$, $S = A \cup \{x_k\}$ with $i \neq k$ uniquely determined. The owner pair of $\tau_{A',i',k'}$ is $(A' \cup \{x_{i'}\}, A' \cup \{x_{k'}\})$, and that of $\sigma_{A',k'}$ is $(A' \cup \{x_{k'+1}\}, A' \cup \{x_{k'}\})$. Both have intersection A', so they equal (T,S) only if $A' = A$, and then only for $(i',k') = (i,k)$, respectively $k' = k$ and $i = k+1$. With $\ell = (k-i) \bmod n \in \{1, \ldots, n-1\}$ this gives $n(T,S) = [\ell \leq \ell_{\max}] + [h \text{ odd},\ \ell = n-1]$. For h even, $\ell_{\max} = n-1$ and there are no side pairs. For h odd, exactly one of $\ell \leq n-2$ and $\ell = n-1$ holds. In both cases $n(T,S) = 1$. Every owner pair meets in exactly two points, so pairs outside Ω^2 have $n = 0$, and side pairs lie in Ω.

Counts: there are $\binom{h}{2}$ axes with n registers of each kind. Since $\binom{h}{2}(h-2) = 3\binom{h}{3}$, there are $3v$ registers of each kind. Each i has $\ell_{\max}$ partners k, and for odd h each k gives one side pair. □

Remark 5.22. For odd h the choice $\ell_{\max} = n-1$ would violate (D3). The last residual of the reader would be $M(Q_{ab})^{\perp}$, which has no ONB by Lemma 5.20(d). The side pairs $\sigma_{A,k}$ replace exactly those transfers.

Coordinate labels: disjoint pairs

For $A \subseteq [h]$ let $\mathbb{F}_2^A = \mathrm{span}\{e_i : i \in A\} \subseteq D$, and write $A^c = [h] \setminus A$.

Lemma 5.23 (coordinate labels and chains)**.** *(a) $\mathbb{F}_2^A$ is nondegenerate with ONB $\{e_i\}_{i\in A}$, $(\mathbb{F}_2^A)^\perp = \mathbb{F}_2^{A^c}$, and for $A \subseteq A'$ the residual of $\mathbb{F}_2^A \subseteq \mathbb{F}_2^{A'}$ is $\mathbb{F}_2^{A'\setminus A}$.*

(b) An X-register with owner T and chain $\mathbb{F}_2^{A_1} \subsetneq \cdots \subsetneq \mathbb{F}_2^{A_k}$ satisfies (D1) and (D2) if $T \subseteq A_1$ and $|A_1| \geq 4$.

(c) A Y-register with owner S and chain $\mathbb{F}_2^{A_1} \subsetneq \cdots \subsetneq \mathbb{F}_2^{A_k}$ satisfies (D1) and (D3) if $S \cap A_k = \emptyset$ and $|A_k| \leq h - 4$.

(d) Suppose $\mathbb{F}_2^A$ lies in the chain of an X-register as in (b), with owner T, and in the chain of a Y-register as in (c), with owner S. Then $T \cap S = \emptyset$; in particular $(T, S) \in \Omega^0$.

Proof. (a) is clear. (b) $t_T \in \mathbb{F}_2^{A_1}$ since $T \subseteq A_1$. The residual $\mathbb{F}_2^{A_1} \cap t_T^\perp$ contains e_i for $i \in A_1 \setminus T$, which exists since $|A_1| > 3$. The residual $\mathbb{F}_2^{A_{i+1}\setminus A_i}$ is nonzero and contains a coordinate unit. Finally $(\mathbb{F}_2^{A_k})^\perp = \mathbb{F}_2^{A_k^c}$ is 0 or contains a coordinate unit. (c) $\mathbb{F}_2^{A_k} \perp t_S$ since the supports are disjoint. The residuals $\mathbb{F}_2^{A_1}$ and $\mathbb{F}_2^{A_{i+1}\setminus A_i}$ are 0 or contain a coordinate unit. The last residual $t_S^\perp \cap \mathbb{F}_2^{A_k^c}$ contains e_x for $x \in A_k^c \setminus S$, which exists since $|A_k^c| \geq 4 > |S|$. (d) Let A_1 be the first set of the X-chain and A' the last set of the Y-chain. Then $T \subseteq A_1 \subseteq A \subseteq A'$ and $S \cap A' = \emptyset$, so $T \cap S = \emptyset$. □

The bounds are sharp. If $|A_1| = 3$ then $A_1 = T$, and $\mathbb{F}_2^T \cap t_T^\perp$ consists of the even-weight vectors of $\mathbb{F}_2^T$, which has no ONB. The case $|A_k| = h - 3$ is symmetric. A coordinate label $\mathbb{F}_2^A$ can serve every writer $T \subseteq A$ and every reader $S \subseteq A^c$.

Definition 5.24 (prefix-chain coordinate design)**.** A *prefix-chain coordinate design* for $D = \mathbb{F}_2^h$ is given by a list of groups $g = 1, \ldots, \Gamma$. Group g consists of the following data.

- A chain of sets $A_{g,1} \subsetneq \cdots \subsetneq A_{g,k_g} \subseteq [h]$ with $4 \leq |A_{g,i}| \leq h - 4$ and $k_g \leq 9$.
- *Writer entries* (T, g, i) with $T \subseteq A_{g,i}$.
- *Reader entries* (S, g, I) with $\emptyset \neq I \subseteq \{1, \ldots, k_g\}$ and $S \cap A_{g,\max I} = \emptyset$.

Its register design $\mathcal{R}^{\text{coord}}$ has the following parts.

- An X-register for each writer entry, with owner T and chain $(\mathbb{F}_2^{A_{g,i}})$, a single label.
- A Y-register for each reader entry, with owner S and chain $(\mathbb{F}_2^{A_{g,i}})_{i\in I}$ in increasing order.
- Candidates $(\mathbb{F}_2^{A_{g,i}}, (T, g, i), (S, g, I))$, one for each writer entry and each reader entry of the same group with $i \in I$. They are ordered lexicographically by (g, i), with ties broken by a fixed order of the entries.
- Transfers and side pairs given by first cover relative to Ω^0 (Lemma 5.9).

In the designs below, $A_{g,i} = \pi_g(\{1, \ldots, s_g + i - 1\})$ for a permutation π_g of $[h]$ and an offset s_g. The sets are prefixes of a permutation, which explains the name. This structure is not used in the proofs.

Lemma 5.25. *$\mathcal{R}^{\text{coord}}$ is a partial register design for Ω^0. Its X-registers have one label, its Y-registers have at most 9 labels, and* $\text{side}(\mathcal{R}^{\text{coord}}) = |\Omega^0| - \#\{\text{covered pairs}\}$.

Proof. Lemma 5.23(b) and (c) give (D1)–(D3), using $A_{g,i} \subseteq A_{g,\max I}$ for $i \in I$. Lemma 5.23(d) shows that every candidate's owner pair lies in Ω^0. Now apply Lemma 5.9. □

Proposition 5.26 (the three searched designs; computational verification)**.** *For $h \in \{30, 20, 41\}$ let $\mathcal{R}_h^{\mathrm{coord}}$ be the prefix-chain coordinate design given by the searched data of Remark 5.27. Let $\mathcal{R}_h = \mathcal{R}_h^{\mathrm{coord}} \sqcup \mathcal{R}_h^{\mathrm{sun}}$.*

(a) The data satisfy the conditions of Definition 5.24. The label sizes lie in $[11, 19]$, $[6, 14]$ and $[16, 24]$ respectively, and the longest Y-chains have 9, 8 and 9 labels.

(b) The numbers of writer entries, reader entries and uncovered disjoint pairs are as in Table 7.

(c) $\mathcal{R}_h$ is a register design, satisfying (D1)–(D5), with the counts of Table 7.

Parts (a) and (b) are established by exhaustive computation, not by a proof by hand; see Remark 5.27 for their formal verification. Part (c) follows from (a), (b), Lemma 5.25, Proposition 5.21 and Lemma 5.8.

	$h = 30$ (stage 1)	$h = 20$ (stage 2)	$h = 41$ (stage 3)
$v = \binom{h}{3}$	4060	1140	10 660
invocations N_0/v	12 152 400	43 279 600	4 628 400
ordered orthogonal pairs $\lvert\Omega\rvert$	12 204 360	833 340	91 143 000
covered by coordinate transfers	11 860 737	772 124	89 880 593
covered by sunflower transfers $3v\ell_{\max}$	328 860	58 140	1 183 260
side pairs (coordinate + sunflower)	14 763 + 0	3 076 + 0	47 167 + 31 980
coordinate X-registers (writer entries)	96 881	18 090	304 414
coordinate Y-registers (reader entries)	94 429	17 552	330 907
sunflower registers $6v$	24 360	6 840	63 960
$\mathrm{regs}(\mathcal{R}_h)$	215 670	42 482	699 281
$\mathrm{side}(\mathcal{R}_h)$	14 763	3 076	79 147
auxiliary wires per invocation $h + \mathrm{side} + \mathrm{regs}$	230 463	45 578	778 469
per bank element $(\mathrm{side} + \mathrm{regs})/v$	56.8	40.0	73.0
OpenAI's $d = \binom{h-3}{3} + 3(h-3)$	3006	731	8550

Table 7: The designs $\mathcal{R}_{30}$, $\mathcal{R}_{20}$, $\mathcal{R}_{41}$.

Remark 5.27 (how Proposition 5.26 was verified). The designs were found by a heuristic search, which has no proof value. They were then checked exhaustively, in exact integer arithmetic, by three independently written programs. Each program checks the conditions of (a) for every group and entry. It also checks that each of the 12 204 360, 833 340 and 91 143 000 ordered orthogonal pairs is covered exactly once, by one transfer or by one pair of the explicit side-pair list stored with the design. Equivalently, the stored list is the set of uncovered pairs, and its length is the count in (b). The three programs agree on all counts in Table 7.

In the formal verification (Section 7) the same group data define partial register designs for Ω^0, with first-cover semantics, and their validity is checked by the Lean kernel itself. The count bounds are proved there: at most 304 414 X-registers, 330 907 Y-registers and 47 167 side pairs at $h = 41$, and likewise at $h = 30, 20$ with the numbers of Table 7. In the formal version each Y-register follows the maximal chain of its group (every label of the group disjoint from its owner). Its edge multiset may therefore differ from that of the searched data. It still satisfies the counts and the chain-length bounds used in Proposition 5.28(d).

5.8 The exponent

Let $\mathcal{N} = \mathcal{N}(\mathcal{R}_{30}, \mathcal{R}_{20}, \mathcal{R}_{41})$ for the stage order $(h_1, h_2, h_3) = (30, 20, 41)$. Then $m = 24\,600$, $N_0 = 49\,338\,744\,000$, and the stage parameters are as follows:

stage j	h_j	(α_j, β_j)	$(\alpha_j - 1)h_j$	$\alpha_j h_j(\beta_j - 1)$	bank edges
1	30	$(1, 820)$	0	24 570	0, 29
2	20	$(30, 41)$	580	24 000	551, 19
3	41	$(600, 1)$	24 559	0	23 960, 40

Let $W^* = 2^{43}$. The *padded network* $\mathcal{N}^*$ adds $W^* - W$ padding roles. Each has the label path $0 \to \langle e \rangle \to \mathbb{D}$, for a coordinate vector e of $\mathbb{D}$, is touched by no gate, and has edges of dimension 1 and $m-1$, with residual ONBs $\{e\}$ and the other coordinate vectors. Its frame ratio is $\Phi_{\mathbb{D}}\Phi_0^{-1} = C^{\otimes m}$ [1, §2.2]. Let $\mathcal{E}$ be the multiset of nonzero residual dimensions of all edges of $\mathcal{N}^*$, and

$$\Phi(z) = \sum_{\rho \in \mathcal{E}} \left(\frac{\rho}{m}\right)^z.$$

Proposition 5.28 (exponent certificate)**.** *For the padded network $\mathcal{N}^*$:*

(a) $W = 8\,475\,019\,577\,600 \le 2^{43} = 8\,796\,093\,022\,208$*, so there are* $321\,073\,444\,608$ *padding roles. Moreover* $\Delta = 26\,618\,807\,200$*, and* $W^*m - \sum_{\rho\in\mathcal{E}} \rho = \Delta > 0$.

(b) $1 \le \rho \le m-1$ *for every* $\rho \in \mathcal{E}$*, and* $|\mathcal{E}| \ge 2W^*$.

(c) $\Phi(1-\varepsilon_1) \le 2^{43}$ *for* $\varepsilon_1 = 3.279039 \cdot 10^{-6}$ *(exact edge multiset).*

(d) $\Phi(1-\varepsilon_2) \le 2^{43}$ *for* $\varepsilon_2 = 3.221860 \cdot 10^{-6}$*, by an argument that uses only the counts of Table 7 and the chain lengths of Lemma 5.25.*

(e) $\Phi(z) < 2^{43}$ *for* $z = 1 - 3.21 \cdot 10^{-6}$.

Proof. (a) Substitute Table 7 into Proposition 5.19(c). Each padding role contributes $1+(m-1) = m$ to $\sum \rho$, so $W^*m - \sum_{\rho\in\mathcal{E}} \rho = Wm - \sum_e \rho_e = \Delta$.

(b) By Table 6 and the stage table above, the fixed sizes are at most 24 570, the register middle edges at most $h_j - 1$, and the padding edges 1 and $m - 1 = 24\,599$. Every role has at least two nonzero edges. A bank wire has at least the stage-1 edge 29 and the stage-2 edges 551 and 19. A centre has three edges h. A side wire has two unit edges. An X-register has the unit edge Bot $\to$ Xb and middle edges summing to $h-1$. A Y-register has the unit edge Yt $\to$ Top and middle edges summing to $h-1$. A padding role has 1 and $m-1$.

(c) Lemma 3.11 bounds each term $(\rho/m)^{1-\varepsilon}$ by a rational number computed from a rational upper bound for $\ln(m/\rho)$. The resulting bound for $\Phi(1-\varepsilon_1)$, evaluated in exact rational arithmetic over the exact multiset $\mathcal{E}$ (38 distinct values of ρ, with the chain dimensions of every register of $\mathcal{R}_h$), is at most 2^{43}. This is a computation, which re-runs the design checks of Remark 5.27 before evaluating the bound. In an independent recomputation of the multiset, the root of $\Phi(p) = 2^{43}$ was found to be $1 - p = 3.2791253 \cdot 10^{-6} \ge \varepsilon_1$.

(d) Put $\varphi(c) = (c/m)^z$ for $c \ge 0$, with $0 < z \le 1$. Since $c \mapsto c^z$ is concave and increasing, Jensen's inequality gives $\sum_{i=1}^n \varphi(c_i) \le n\,\varphi(S/n)$ whenever $c_i \ge 0$ and $\sum c_i \le S$ (tangent bound). For integers $c \ge 1$ also $\varphi(c) \le c\,\varphi(1)$ (crude bound). The *middle edges* of an X-register are its edges from the step-4 touch at Xb(t_T) to the step-10 touch at Top. The middle edges of a Y-register are its edges from the step-3 touch at Bot to the step-8 touch at Yt(t_S). A register with k chain labels has $k+1$ middle edges, and by Table 5 their dimensions sum to $h-1$. Its other nonzero edges are $(\alpha-1)h$, one unit edge and $\alpha h(\beta-1)$. By Lemma 5.25, a coordinate X-register has 2 middle edges

and a coordinate Y-register at most 10. Bounding the coordinate registers by the tangent bound and the sunflower registers by the crude bound gives, with $a = \alpha_j$, $b = \beta_j$ and $h = h_j$ at stage j,

$$\begin{aligned}
\Phi(z) \le{}& \sum_{j=1}^{3} \frac{N_0}{v_j}\Big[\,2v_j\,\varphi_{\mathrm{bk}} + h_j\,\varphi_{\mathrm{c}} + \mathrm{side}_j\,\varphi_{\mathrm{sd}} + X_j\,\varphi_X + Y_j\,\varphi_Y + 6v_j\,\varphi_{\mathrm{sun}}\Big] \\
&+ (W^* - W)\big(\varphi(1) + \varphi(m-1)\big), \\
\varphi_{\mathrm{bk}} &= \varphi((a-1)(h-1)) + \varphi(h-1), \qquad \varphi_{\mathrm{c}} = \varphi((a-1)h) + 3\varphi(h) + \varphi(ah(b-1)), \\
\varphi_{\mathrm{sd}} &= \varphi((a-1)h) + 2\varphi(1) + \varphi(h-2) + \varphi(ah(b-1)), \\
\varphi_X &= \varphi((a-1)h) + \varphi(1) + 2\varphi(\tfrac{h-1}{2}) + \varphi(ah(b-1)), \\
\varphi_Y &= \varphi((a-1)h) + \varphi(1) + 10\,\varphi(\tfrac{h-1}{10}) + \varphi(ah(b-1)), \\
\varphi_{\mathrm{sun}} &= \varphi((a-1)h) + h\,\varphi(1) + \varphi(ah(b-1)),
\end{aligned}$$

where $\varphi(0) = 0$, and X_j, Y_j, side_j are the coordinate X-registers, coordinate Y-registers and side wires of Table 7. Each φ-value at $z = 1 - \varepsilon_2$ is bounded by Lemma 3.11 with rational logarithm bounds, 22 distinct arguments in all. The resulting rational number is at most 2^{43} (exact arithmetic).

(e) Each term of Φ is strictly decreasing in z, since $0 < \rho/m < 1$. As $1 - 3.21 \cdot 10^{-6} > 1 - \varepsilon_2$, part (e) follows from (d), and also from (c). The rational inequality obtained directly from the bound of (d) at $z = 1 - 3.21 \cdot 10^{-6}$ is part of the formal verification (Section 7). □

Remark 5.29. The argument of part (d) uses only the counts of Table 7 and the chain-length bounds. It therefore applies to any designs whose per-factor counts and chain lengths are dominated by these, and gives for them $W \le 2^{43}$, $1 \le \rho_e \le m - 1$ for every edge, and $\sum_e (\rho_e/m)^z < 2^{43}$ at $z = 1 - 3.21 \cdot 10^{-6}$.

For comparison, the per-direction rule of [1, Theorem 2.6], with the same packing into 2^{43} roles, gives only $1 - \theta \approx \Delta/(W^* m \ln m) \approx 1.2167 \cdot 10^{-8}$ for this network. Almost all of $\sum \rho$ lies in edges with ρ close to m, which the grouped recursion charges nearly nothing.

5.9 The colour-register theorem

Theorem 5.30 (colour-register network)**.** *Let $(h_1, h_2, h_3) = (30, 20, 41)$, let $\mathcal{R}_{30}$, $\mathcal{R}_{20}$, $\mathcal{R}_{41}$ be the register designs of Proposition 5.26, and let $\mathcal{N}^*$ be the network $\mathcal{N}(\mathcal{R}_{30}, \mathcal{R}_{20}, \mathcal{R}_{41})$ padded to $W^* = 2^{43}$ roles. Then $\mathcal{N}^*$ satisfies the hypotheses of Theorem 3.7 with $W^* = 2^{43}$ and $m = 24\,600$:*

(i) $\mathcal{N}^*$ is a network on W^* roles that satisfies the network interface (2.4). For arbitrary contents of all roles and every $f \ge 1$, it applies $C^{\otimes mf}$ to every role.

(ii) Every edge has nested nondegenerate endpoint labels, a residual with an ONB, and residual dimension $1 \le \rho_e \le m - 1$.

(iii) $\displaystyle\sum_e \Big(\frac{\rho_e}{m}\Big)^z < 2^{43}$ *for $z = 1 - 3.21 \cdot 10^{-6}$.*

The network has $W = 8\,475\,019\,577\,600$ non-padding roles and saving $\Delta = 26\,618\,807\,200$. Inequality (iii) also holds with $z = 1 - 3.279039 \cdot 10^{-6}$, as $\le$. Consequently, Theorem 3.7 applies to $\mathcal{N}^*$ with exponent $z = 1 - 3.21 \cdot 10^{-6}$.

Proof. By Proposition 5.26, each $\mathcal{R}_h$ is a register design. Part (i) is Proposition 5.19 for $\mathcal{N}$. The padding roles are touched by no gate, and their frame ratio is $C^{\otimes m}$. Part (ii) is Proposition 5.19(b)

together with Proposition 5.28(b) and the description of the padding roles. Part (iii) and the remaining numbers are Proposition 5.28(a), (c) and (e). □

Remark 5.31 (status of the proof). The proof of Theorem 5.30 has two kinds of input.

(1) *Written proofs* in this section, for general parameters: Lemmas 5.3, 5.6, 5.8, 5.9, 5.13, 5.14, 5.16, 5.20, 5.23, 5.25 and Propositions 5.19, 5.21. They rely on [1, §2] (Lemmas 2.1–2.3, Proposition 2.4 and the splitting statement at the end of §2.3), on standard facts about symmetric bilinear forms over $\mathbb{F}_2$, and on Lemmas 3.1 and 3.11.

(2) *Exhaustive computation* on the three searched coordinate designs (Proposition 5.26(a),(b)), done by independent programs, and the exact rational evaluation of the certificate bounds (Proposition 5.28(c),(d)).

In addition, Theorem 5.30, and with it Theorem 1.1, has been formally verified (Section 7). The formal proof covers the network for arbitrary register designs, the three factor designs, whose data are checked by the Lean kernel, and the exponent inequality in the form of Proposition 5.28(d) at $z = 1 - 3.21 \cdot 10^{-6}$.

6 Proof of the Main Theorem

Proof of Theorem 1.1. By Theorem 5.30, the padded colour-register network with factor sizes $(30, 20, 41)$ and the designs of Proposition 5.26 is a $(24600, 2^{43})$-certificate whose edge sizes lie in $[1, 24599]$ and which satisfies (2) for $z = 1 - 3.21 \cdot 10^{-6}$. Theorem 3.7 gives $C^{\otimes k}$ in $O(2^k(k+1)^z)$, Proposition 2.1 gives $O\big(n(\log n)^z(\log\log n)^{4-z}\big)$, and Corollary 2.2 with $q = 1 - 3.2 \cdot 10^{-6} > z$ gives $O(n(\log n)^q)$. □

7 Formal verification

Theorem 1.1 and Theorem 1.2 have been formally verified in the proof assistant Lean 4 with the Mathlib library. The formal proof extends OpenAI's published Lean formalization [2] of [1].

The formal statement is phrased in OpenAI's formal RAM model, with its operation-count semantics. It asserts that there exist a program computing the root order D from n and a program computing $F_n x$ exactly, for every $n \geq 1$ and $x \in \mathbb{C}^n$, whose total work $W(n)$ satisfies OpenAI's original bounds and

$$W(n) = O\Big(n\,(\log n)^{1-3.2\cdot 10^{-6}}\Big),$$

and whose integers, addresses and storage are polynomially bounded. The formal version of Theorem 1.2 is the same statement with the exponent $1 - 10^{-8}$. Both statements are also verified in their analogues for exact convolution.

The proof uses only the standard axioms of Lean's foundations (propositional extensionality, the axiom of choice and the soundness of quotients) and no unverified computation. In particular, the finite design data of Proposition 5.26, about $1.04 \cdot 10^8$ ordered pairs, are checked by the Lean kernel itself (Remark 5.27).

OpenAI's definitions of the machine model and of correctness are used unchanged, and so is the rest of their formal proof, in particular the reduction of Proposition 2.1, which the kernel re-checks. Only three of OpenAI's files were modified, at the interface between the network and the recursion, where the per-direction recursion is replaced by the grouped one. The new material consists of

the grouped recursion of Section 3, the colour-register network of Section 5, the designs, and the exponent inequality.

The formal proof differs from the written one in two respects. First, it certifies the rate inequality (2) at $z = 1 - 3.21 \cdot 10^{-6}$, through the bound of Proposition 5.28(d), and not near the exact root $1 - p = 3.2791 \cdot 10^{-6}$. Second, it obtains the slightly stronger bound $O\big(n(\log n)^z(\log\log n)^2\big)$, which implies Theorem 1.1 by Corollary 2.2.

What must be trusted is the Lean kernel, and the claim that OpenAI's formal model faithfully expresses exact computation and its cost. The formal development is available from the author.

8 Summary and outlook

We improved the exponent of OpenAI's exact power-saving DFT in two independent ways.

1. The *grouped recursion* (Theorem 3.7) runs each edge of a network as one tensor power. It replaces $\log_m(\sum \rho_e/W^*)$ by the root of $\sum_e(\rho_e/m)^p = W^*$, never worse and typically much better. On OpenAI's formalized network it gives the bound $O\big(n(\log n)^{1-10^{-8}}\big)$ (Theorem 1.2), against the previously formalized $1 - 2 \cdot 10^{-11}$.

2. *Colour registers* (Section 5) replace the side wires of the paper's network by shared registers meeting at common labels. Combined with the grouped recursion they give $O\big(n(\log n)^{1-3.2\cdot 10^{-6}}\big)$ (Theorem 1.1), seven orders of magnitude beyond the original gap of $2.1 \cdot 10^{-13}$.

Both results have been formally verified (Section 7).

Open problems.

- *Better register designs.* The searched coordinate designs are about three times above a simple counting lower bound. Designs in which each element needs $O(1)$ registers ("flag-meeting" designs) would roughly triple $1 - p$. We do not know whether they exist.

- *Lorentzian families.* Replacing triples by the Lorentzian vector families of our earlier analysis gives $1 - p \geq 2.479 \cdot 10^{-7}$ with side wires. Combining them with colour registers requires meeting labels for pair-type vectors, which we have not constructed.

- *Limits of the method.* Within the 12-step register pattern, the stage-2 auxiliaries dominate the cost, because their labels must contain $B \otimes D_2$. A different invocation pattern would be needed to go substantially further.

Acknowledgements

This work builds on OpenAI's paper and on their open Lean formalization.

References

[1] OpenAI, *An explicit power saving for the exact discrete Fourier transform*, September 2026.

[2] OpenAI, *Lean formalization of* [1], `https://github.com/openai/math`.